

Negation causes NPI and NCI illusions in Czech

Radim Lacina^{1,2} and Mojmír Dočekal¹

¹Department of Linguistics and Baltic Languages, Faculty of Arts, Masaryk University

²Department of Czech Language, Faculty of Arts, Masaryk University

Submitted on February 27, 2026

Corresponding author:

Radim Lacina
rlacina@mail.muni.cz
Arne Nováka 1/1
602 00, Brno
Czech Republic

Negation causes NPI and NCI illusions in Czech

Abstract

Grammaticality illusions are cases where comprehenders fleetingly consider ungrammatical sentences to be acceptable. The phenomenon been used to tap into the mechanism of online negative polarity item (NPI) licensing (Muller & Phillips, 2020). The effect has high specificity and not all environments that ordinarily license NPIs create illusions—sentential negation being the prime case that is ineffective as an illusory licenser (Orth et al., 2021). In the current study, we contrast NPIs with another class of expressions, namely negative concord items (NCIs), which are said to be licensed in the syntax unlike NPIs. We test whether sentential negation gives rise to illusions with Czech NPIs and NCIs respectively, predicting illusions for NCIs and no effect for NPIs. However, in three speeded acceptability judgement experiments, we find illusions with both types of expressions. Our data are inconsistent with the active scope (Orth et al., 2021) and scalar theories (Muller, 2022; Schwab, 2023). The cue-based retrieval (Vasishth et al., 2008) and the feature percolation approaches (Eberhard et al., 2005) are both compatible with the results. We argue in favour of feature percolation as the best explanation for both our Czech data as well as the data from other languages, appealing to cross-linguistic differences in the placement of negation within syntactic structure. While negation is high up in the syntactic structure in Czech and can therefore percolate up and cause illusions, it is low in English, not allowing for illusions to arise. This result highlights the importance of language-specific syntactic structure for online comprehension.

Keywords: negative polarity items; negative concord; grammaticality illusions; Czech; speeded acceptability judgements

1 Introduction

Many languages of the world have lexical items whose grammaticality depends in one way or another on the presence of negation (or a negative-like environment), chief among these are negative polarity items (NPIs) and negative concord items (NCIs, also referred to as n-words or neg-words) (Giannakidou & Zeijlstra, 2017). This licensing requirement presents a challenge in online language processing, as comprehenders must check upon encountering an NPI or an NCI whether their licensing conditions are met. These two types of negative dependency have been proposed to be underpinned by different mechanisms. While NPIs are largely seen as semantically or pragmatically dependent (Barker, 2018; Crnić, 2011) expressions, NCIs are analysed as a type of syntactic agreement (Zeijlstra, 2004, 2008). Both of these types of expressions have been found to be susceptible to selective *grammaticality illusions* during comprehension caused by illusory licensors (see Vasishth et al., 2008; Lacina, 2025, for NPIs and NCIs respectively). These are expressions that could ordinarily license either NPIs or NCIs, but are not in the structurally appropriate position within a sentence to facilitate proper licensing. Nevertheless, their presence causes comprehenders to temporarily view unlicensed NPIs and NCIs as acceptable. Crucially, the precise conditions that give rise to these illusions effects have been used to test theories of the online licensing of these expressions (Phillips et al., 2011; Muller & Phillips, 2020).

In the current study, we ask whether the online processing of these two seemingly similar negation-dependent expressions is underpinned by the same mechanisms or not. We do this by testing and contrasting grammaticality illusions with NPIs and NCIs in a language that contains both of these expressions—Czech. We derive different predictions for the two types of expressions based on the respective semantic-pragmatic or syntactic nature of their licensing. This difference predicts that under one specific condition, namely sentential negation as an illusory licensor, illusions should not occur with Czech NPIs, but should occur with NCIs. Testing these predictions will ultimately give us insight into how these negatively dependent expressions are processed by comprehenders in an ongoing stream of linguistic signals and whether this processing is the same for NPIs and NCIs.

Below, we begin with a brief introduction of the theoretical treatments of the licensing conditions of NPIs and NCIs with a focus on Czech. We will argue that while NPIs are semantically and pragmatically dependent expressions, NCIs function akin to syntactic agreement. We then discuss the phenomenon of grammatical illusions with special attention paid to NPI and NCI illusions as well as the related agreement attraction effect. We will argue that since negative concord is analysed as agreement, the NCI illusion could be treated as a type of agreement attraction. At this point, we introduce the major theories that have been proposed to explain both NPI illusions and agreement attraction—cue-based retrieval and feature percolation. Then we discuss approaches that decouple the two effects and propose explanations specific to NPI illusions—the scalar ap-

proach and the active scope approach. We will focus specifically on how these theories deal with the recent empirical finding that while sentential negation is a valid licenser of NPIs, it is not an illusory licenser (Orth et al., 2021; Muller, 2022, among others). From this, we will derive our predictions for the current study, which are that while for Czech NPIs, sentential negation should be ineffective as an illusory licenser, for Czech NCIs, sentential negation should cause illusions due to the syntactic nature of their licensing. We then report three speeded acceptability judgement experiments testing these predictions. Contrary to our expectations, we find strong evidence for illusory licensing with both NPIs and NCIs as caused by sentential negation. In the discussion, we argue that this goes directly against the scalar and active scope theories. We put forward that the feature percolation account fits our Czech data as well as the earlier English findings best, since it can accommodate the structural differences between the two languages in terms of the expression of negation.

1.1 NPIs and NCIs

Negative polarity items occur prolifically in many languages of the world and “seem to exist in virtually every language we consider” (Giannakidou, 2019, p. 70). They are licensed in only certain environments, such as within the scope of a downward-entailing operator (Ladusaw, 1992), exemplified with the NPI *ever* in (1-a) and (1-b):

- (1) a. No student has *ever* got an A.
 b. *The student has *ever* got an A.

NCIs (negative concord items; also referred to as n-words or neg-words), on the other hand, are found only in languages with negative concord (e.g., Giannakidou, 2000) in which expressions corresponding to for example *nobody*, *never* or the negative quantifier *no* require the presence of sentential negation within the same clause, while yielding only one logical negation—the Czech example (2-a) is equivalent to the English sentence *No student got an A*. The negative quantifier *žádný* (‘no’) present in a sentence without the sentential negation morpheme *ne-* attached to the verb is ungrammatical, as illustrated in (2-b).

- (2) a. Žádný student ne-dostal za A.
 No student NEG-got for A.
 Lit. ‘No student didn’t get an A.’
 b. *Žádný student dostal za A.
 No student got for A.
 Lit. ‘No student got an A.’

A language, Czech included, may have both of these types of negatively dependent expressions. However, the distribution and licensing conditions of NPIs and NCIs differ even within a single language. Most theories of NPIs treat these expressions as semantically and/or pragmatically

dependent. So far, the arguably most influential theory of NPIs follows Ladusaw (1992) in his identification of downward-entailing (DE) contexts as the licensing environments for NPIs (but see von Stechow (1999) for some issues). Other trends identify scope marking as the crucial semantic condition (Barker, 2018) or claim that NPIs are licensed by the computation of scalar implicatures (Crnič, 2011; Chierchia, 2013, among others). On the other hand, arguably the most subscribed-to theory of NCIs claims that these expressions are licensed in the syntax in a process akin to agreement (Zeijlstra, 2004, 2022; Penka, 2011).¹ In this theory, NCIs bear an uninterpretable lexical feature [uNeg]. The [uNeg] feature is a formal requirement representing the syntactic sensitivity of NCIs, since it must agree with an interpretable negative feature [iNeg] coming from sentential negation operator $\neg$ Op. The agreement is subject to the usual locality constraints, which govern other kinds of agreement (subject-verb agreement, modifier-head agreement, etc.) in the particular language. Thus, while the licensing of NPIs is semantic/pragmatic, it is only syntactically dependent for NCIs.

In Czech, this is evidenced by the distributional difference between the negative quantifier NCI *žádný* and the NPI *ani* ('not even'). While the NCI strictly requires sentential negation within its local clause to be grammatical, the NPI can be licensed outside of it with neg-raising predicates. These are verbs such as *věřit* ('believe') which, when negated, are interpreted as if the negation is raised to the higher clause from its embedded clause. Czech speakers accept NPIs in the embedded position of neg-raising predicates and reject NCIs in the same position, as seen in (3) (Dočekal & Dotlačil, 2016, 2018). Pragmatically, the negation interpreted on the embedded predicate suffices to license NPIs but not NCIs.

- (3) a. Petr ne-věří, že by to ani jeden z nich zvládl.
 Petr NEG-believes that would it not.even one of them manage
 'Petr doesn't believe that even one of them would manage.'
- b. *Petr ne-věří, že by to žádný z nich zvládl.
 Petr NEG-believes that would it no of them manage
 'Petr doesn't believe that none of them would manage.'

The second experimental finding is that Czech speakers accept the NPI *ani* ('not even') only in certain contexts with the correct probability manipulation (Dočekal & Dotlačil, 2016, 2018). When judging sentences such as (4-a), they accepted the NPI *ani* only when the associated NP denoted a low scalar point of an ad-hoc scale. Not becoming a priest is less likely than not becoming a bishop or a pope. The sentence with the NCI was judged to be acceptable in the same context with both ends of the scale, (4-b). The experimental results suggest that Czech NPIs are sensitive to the likelihood of the prejacent, while NCIs are not.

- (4) a. Petr se ne-stal ani knězem/#ani papežem.
 Petr REFL NEG-became even priest/even pope
 'Petr didn't even become a priest/a pope.'

¹But see Section 4.2 for an alternative approach treating NCIs and negative concord semantically.

- b. Petr se ne-stal žádným knězem/žádným papežem.
 Petr REFL NEG-became no priest/no pope
 ‘Petr didn’t become a priest/a pope.’

The upshot of these findings for the purposes of the current study is that the licensing conditions for NPIs and NCIs differ in Czech. While for NCIs, licensing is a type of syntactic agreement and, therefore, subject to locality constraints, NPIs have a more substantial meaning component to their licensing. We argue that this could be reflected in the online processing of these expressions. The way we target processing is by exploiting the phenomenon of grammaticality illusions, discussed next.

1.2 Grammaticality illusions

The section above describes the licensing conditions for NPIs and NCIs, i.e., situations where all goes well. However, it has been observed that in online processing, comprehenders can sometimes accept NPIs or NCIs even when they are unlicensed. This happens under the condition that a specific distractor is present. Such cases are known as *grammaticality illusions*—situations in which comprehenders are led astray and judge ungrammatical sentences as nevertheless acceptable (Phillips et al., 2011). Let us start with an example of the NPI illusion in English:

- (5) *The student that *no teacher* praises has *ever* got an A.

Sentences such as (5) are readily accepted by native speakers of English even though they are ungrammatical, since there is no licensor in the matrix clause. The key lies in the *illusory licensor* in the relative clause (RC), the negative quantifier *no*. While this quantifier does not scope over the NPI *ever* and thus cannot properly license the expression, it has been found to nevertheless increase comprehenders’ acceptance rates of sentences such as (5) when compared to those where the negative quantifier is not present in the RC as in (6) below. This effect has been repeatedly found both in English and in many other languages (Drenhaus et al., 2005; Vasishth et al., 2008; Parker & Phillips, 2016; Ng & Husband, 2017; Mendia et al., 2018; Yun et al., 2018; Schwab, 2023; Muller, 2022; Orth et al., 2021, 2025; Su & Aparicio, 2022; Bakay & Oğuz, 2024; Thorsen, 2025; Kush et al., 2026).

- (6) The student that *the teacher* praises has *ever* got an A.

This increase in perceived acceptability is, however, fleeting, as this illusion is typically only seen when comprehenders are under time pressure to make their judgement or in other measures tapping into early processing (Muller & Phillips, 2020). The NPI illusion has also been found to surface in reading times (Xiang et al., 2013; Orth et al., 2025) and in event-related potentials (ERPs) (Xiang et al., 2009; Yanilmaz & Drury, 2018). Typically, when participants perform the *untimed* acceptability judgement task, sentences such as (5) no longer show any increased acceptability

(Phillips et al., 2011; Muller & Phillips, 2020) when compared to sentences like (6).

The same illusion pattern was recently found for NCIs in Czech by Lacina (2025). He tested whether Czech comprehenders came under grammaticality illusions when an unlicensed NCI quantifier *žádný* (‘no’) in a matrix clause was accompanied by a licensed instance of the same NCI in an intervening RC:

- (7) *Minulý měsíc vyhrál karbaník, který ne-hledá žádnou práci, žádnou sázku.
Last month won gambler who NEG-seeks no work, no bet
‘Last month, the gambler, who is looking for no work, won no bet.’

He found that his participants judged sentences such as (7) as acceptable more often than sentences where the RC included neither the negative quantifier nor the negation. This difference was only seen when the participants were under a time limit to make their judgements. No differences due to the contents of the RC were detected in an untimed acceptability judgement experiment.

This time-sensitivity and presence in early processing suggest that these effects are in fact an illusion—an appearance of grammatical well-formedness—rather than a structure that is generated by the grammar. The reason why much research attention has been given to linguistic illusions is that the selectivity in under what conditions they arise can be used to better understand on-line language processing, as has previously proven effective in other domains of cognitive science (Phillips et al., 2011)², such as, prominently, vision (see Todorović, 2020, for an in-depth discussion). When and how a system fails can instruct us about what makes it run when functioning properly. Recent attempts have been made to explain illusions as a broad phenomenon caused by comprehenders aiming to rationally infer intended messages (Zhang, 2024) or as the result of such inference making in combination and competition with good-enough processing (Paape, 2024).

As far as language processing is concerned, illusions are also informative with regards to two questions. Firstly, looking within a single language, the specific conditions under which they arise give a glimpse into of what representations and processes are used and when. Secondly, expanding investigations to other languages informs us not only about the generalisability of the observed effects, but crucially about how the specific features of languages influence how comprehenders encode, store, retrieve and use linguistic structures.

In Section 1.1, we made the claim that while NPI licensing is semantic/pragmatic in nature, NCI licensing is syntactic agreement. Agreement is well known to be susceptible to the illusion effect known as agreement attraction (see Parker, 2025b, for a recent in-depth treatment of the phenomenon). This refers to cases with an erroneous match of morphosyntactic features such as number or gender with a distractor element that is not the true controller (for seminal studies in production and comprehension respectively, see Bock & Miller, 1991; Wagers et al., 2009). Take

²Other linguistic illusion besides NPI illusions and agreement attraction discussed prominently in the current study, include the comparative illusion (O’Connor et al., 2013; O’Connor, 2015; Wellwood et al., 2018), forwards anaphora (Badecker & Straub, 2002; Runner et al., 2006), fallibility in case licensing (Bader et al., 2000), the missing-XP effect (Gibson & Thomas, 1999; Häussler & Bader, 2015; N. Huang & Phillips, 2021; Pañeda & Lago, 2024) and the missing quantifier illusion with French *de* (Pasquereau et al., 2024).

the following example:

- (8) *The cost of the improvements have not yet been estimated.

Sentence (8) is ungrammatical, since the auxiliary *have* mismatches the subject head *the cost* in number. Rather, it appears to match the number of the linearly closer phrase *the improvements*, which is, however, structurally irrelevant for agreement. When exposed to sentences such as (8), comprehenders come under the illusion that said sentence is grammatical (Wagers et al., 2009), similar to cases of unlicensed NPIs. In comprehension, this effect has been recorded in many languages both in speeded acceptability judgements (Wagers et al., 2009; Patson & Husband, 2016; Hammerly et al., 2019; Royer, 2021; Lacina & Dotlačil, 2024; Paspali & Marinis, 2020; Gonzalez Alonso et al., 2021), and in other online measures such as sped-up reading times (Wagers et al., 2009; Lago et al., 2015; Parker & An, 2018; Tucker et al., 2015, 2021; Avetisyan et al., 2020; Slioussar & Malko, 2016; Slioussar, 2018; Chromý et al., 2023a; Lacina et al., 2025; Kwon & Sturt, 2016; Sturt & Kwon, 2024; Franck et al., 2015; Lago et al., 2019; Turk & Logačev, 2021; Ivanova-Sullivan et al., 2024).

It is plausible to suggest that we ought to treat the NCI illusion as a case of agreement attraction. This matters, as the two phenomena, NPI illusions and agreement attraction, have been the subject of debate with regards to both their similarities and their differences. On the one hand, two influential theories, the cue-based retrieval model (Lewis & Vasishth, 2005; Vasishth et al., 2008; Engelmann et al., 2019) and the feature percolation account (Eberhard et al., 2005), propose a joint explanation for both agreement attraction and NPI illusions. On the other hand, the two types illusions have been found to behave differently, giving rise to two prominent NPI-specific accounts, the active scope theory (Orth et al., 2021) and the scalar alternative approach (Schwab, 2023; Muller, 2022). Below, we discuss these theories as well as the empirical landscapes connecting and differentiating NPI illusions and agreement attraction. We will focus specifically on the efficacy of sentential negation as an illusory licenser, which will serve as the main predicted factor of a difference between Czech NPI and NCI illusions in the current study.

1.3 The theoretical and empirical landscape of NPI illusions and agreement attraction

1.3.1 Cue-based retrieval and feature percolation

The feature percolation approach, inclusive of the the Marking and Morphing theory, originates in attempts at explaining attraction effects in language production (Eberhard et al., 2005), that is cases of speech errors where speakers produce sentences with faulty agreement such as the one in (8) and has also been applied to comprehension. The explanation proposed by feature percolation accounts for both agreement attraction and NPI illusions is that features (such as number) can

percolate up the syntactic tree out of their original placement. Phrases are taken to be able to take feature values ranging from -1 to 1 (e.g., from singular to plural), which are the result of all the features found within a particular phrase. Take the example in (8). Here, the overall value of the phrase *the cost of the improvements* would not be fully singular (say -1), since the plurality of *the improvements* would drag the value in the other direction. As a consequence of this, the whole phrase might be misjudged to be plural.

The same approach, Orth et al. (2021) say, could be applied to the case of NPI illusions with polarity as the relevant feature. Thus, in the example (5) (**The student, who no teacher praises, has ever got an A*), the negative quantifier influences the polarity value outside of the relative clause. Additionally, these accounts may be specified in such a way that the place of a particular feature-carrying element within the syntactic structure could be relevant for the strength or even appearance of illusion effects (Orth et al., 2021). The idea is that the deeper the attractor or illusory licenser is embedded within the structure, the less it is capable of influencing the eventual feature value of the whole. Thus, structurally deeper elements are predicted to be less effective at eliciting agreement attraction and NPI illusions. In recent work, Orth et al. (2021) argues that this prediction is not borne out fully. In their study, they found that while structurally higher elements were capable of causing NPI illusions, the account struggled to explain the behaviour of more embedded illusory licensors. Curiously, while medially placed illusory licensors such as sentential negation failed to elicit illusions, even more deeply embedded ones succeeded.

The cue-based retrieval approach is based on the idea that during online language processing, comprehenders encode each piece of incoming linguistic material as it comes to them linearly as the sentence in question unfolds in time (Lewis & Vasishth, 2005). These encodings take the form of “bundles of features”, which include, among others, grammatical gender, number and subjecthood. These are then stored for the course of comprehending a sentence. Next, at a point when the comprehender needs to establish a long-distance dependency, for example subject-verb number agreement, the element in need of dependency sends out a probe specifying what element ought to be retrieved from memory based on a certain cues. For example, in the agreement attraction sentence (8), the verb *have* requires agreement in number with the subject. The probe would then directly access all stored elements and search for those that match the “+plural”, “+subject” features. The way that agreement attraction or NPI illusion effects are explained within this framework, then, is by appealing to misretrieval due to a partial match of a particular distractor. When a sentence is ungrammatical, i.e., there is no perfect match for the cues, yet a partial match with an element that is not the true controller/licensor can be established, the theory predicts that this distractor may be selected and that processing may be sped up when compared to ungrammatical sentences with no such partially matching distractor (Logačev & Vasishth, 2016).

Similarly to feature percolation, the cue-based retrieval theory applies the same story to both agreement attraction and NPI illusions. The only difference lies in the types of features and

cues that are encoded and retrieved by comprehenders in the course of language processing. In the case of NPI illusions, [Vasishth et al. \(2008\)](#) propose that the cues that the NPI requires are “+c-commanding” and “+negative”. In sentences where there is no licenser in the matrix clause, i.e., not a single element that would match both the features, yet one that has the latter feature (“+negative”) without the former (“+c-commanding”), a partial-match configuration results. It is precisely in this situation where the illusory licenser (for example the often used quantifier *no*) is misretrieved and the NPI in the matrix clause is erroneously licensed.

1.3.2 Empirical issues for joint accounts

We have now covered two major accounts that have been invoked to explain both NPI illusions and agreement attraction. This has consequences as to what predictions these approaches make. In this section, we focus on these predictions and on how they have fared empirically. Firstly, the general prediction is that if the explanation for NPI illusions and agreement attraction is the same, the two phenomena should not be differently affected by the same conditions or manipulations. Next, for NPI illusions, the cue-based retrieval theory predicts is that under all circumstances in which there is a distractor that has the “+negative” feature and there is no perfect match, the effect should occur. This means that illusions should arise for all potential licensors and NPIs that share the same licensing conditions. The feature percolation account makes the prediction that all NPIs should be susceptible to illusions. On the assumption that embedding depth matters, the feature percolation account can accommodate cases where not all licensors can also be effective illusory ones, that is provided they are too deeply embedded to percolate up.

Let us start with the first claim of the fundamental similarity of NPI illusions and agreement attraction. A crucial difference between them has been found to lie in the influence of distance. [Parker & Phillips \(2016\)](#) tested the effect of increasing the distance between the attractor or illusory licenser and the agreeing verb or NPI respectively within the illusion-inducing configurations discussed above. What they found was that while in the case of subject-verb agreement attraction, inserting extra material did not have the effect of turning the illusion off, the same manipulation caused comprehenders to no longer judge unlicensed NPIs as acceptable. Furthermore, in her dissertation, Hanna Muller ([2022](#)) recently showed that in the case of NPI illusions, it is not the distance from the illusory licenser itself that matters; rather it is the distance from the edge of the scopal boundary of the licenser. This in turn suggests that unlike in agreement attractions, the appearance of NPI illusions is connected not to the presence of a licenser but rather an environment capable of licensing these expressions, as argued in the theoretical semantics literature (since [Ladusaw, 1992](#), already). This went against the tenet of the cue-based retrieval theory that did not predict any differences between agreement attraction and NPI illusions. On the other hand, this could potentially be explained within the feature percolation account, depending on how distance is achieved, since the intervening material could make the illusory licenser more deeply embedded

and incapable of having the negative polarity percolate up.

We now turn to the other issue—whether all NPI illusions are made alike. As it turns out, NPI illusions have been found to be highly selective with regards to the contexts in which they arise, as it is not the case that the phenomenon occurs with *all* possible licensors and *all* NPIs, contrary to the predictions of the cue-based retrieval model.

Orth et al. (2021) investigated whether all potential NPI licensors behave the same with regards to illusions. Take the contrast between the following two sentences:

(9) The senator who *no* farmer believed to be untrustworthy will *ever* vote yes on the smoking proposal.

(10) The senator who *didn't* believe the farmer to be untrustworthy will *ever* vote yes on the smoking proposal.

What these researchers found was that while illusions were observed when the RC contained the negative quantifier *no* as in (9), this did not occur with simple sentential negation in (10). Similarly, de Dios Flores et al. (2017) found that *haven't* likewise cannot serve as an illusory licensor. In her dissertation, Muller (2022) also replicated the finding that in English, sentential negation does not give rise to NPI illusions. Orth et al. (2021) interpreted these results as going against the predictions of the cue-based retrieval approach to illusory NPI licensing, as the account predicts that all licensors should give rise to illusions with all NPIs.

Interestingly, however, sentential negation was found to be effective in giving rise to NPI illusions in another language, namely Turkish. Yanilmaz & Drury (2018) exposed their participants to sentences with an unlicensed Turkish NPIs such as *kimse* ('anybody') while recording the their participants' immediate neural responses with EEG. Unlike previous studies on English (Parker & An, 2018; Muller, 2022) and German (Drenhaus et al., 2005; Schwab, 2023), Yanilmaz & Drury (2018) studied *prospective* NPI licensing, meaning that the NPI preceded the licensor. They found that when negation was present in an embedded RC with the matrix verb following in the affirmative, Turkish comprehenders exhibited patterns of neural response in line with the interpretation that they came under a grammaticality illusion. Recently, Bakay & Oğuz (2024) supported these results by reporting that Turkish comprehenders show grammaticality illusions in their speeded acceptability judgements of sentences with illusory licensors.

While this finding might *prima facie* be incompatible with the data showing sentential negation to be ineffective in causing illusions (de Dios Flores et al., 2017; Orth et al., 2021; Muller, 2022), there is a way to accommodate it, as argued by Muller (2022). In her view, the key lies in the different word order of English and Turkish sentences. In the head-final Turkish, comprehenders

encounter the NPI first and consequently produce a strong prediction that a licenser is coming. This is then already satisfied at the point of encountering the negation in the embedded relative clause. This then ameliorates the effect of no negation in the matrix clause. The same reasoning could then be applied to the findings of Yun et al. (2018) and of Su & Aparicio (2022), who found sentential negation to be effective in creating NPI illusions in the similarly head-final Korean and Japanese respectively.³

Finally, there is the prediction that all NPIs, regardless of their type, ought to be susceptible to illusory licensing. This prediction has also not turned out to be correct. Parker & Phillips (2016) tested two English NPIs for their illusion susceptibility, namely *ever* and *any*. In their acceptability study, they found that only the former was able to elicit illusions in their participants. However, the recent study by Kush et al. (2026) contradicts these results. In a high-powered study, they found comparable illusion effects both with *ever* and with *any* under the condition that the linear position of the two NPIs is controlled for. However, differences between different NPIs have still been reported in the literature. Schwab (2023) set out to test NPI illusions with two types of expressions—strengthening NPIs such as *ever* and attenuating ones. In her study, she showed that in German, only strengthening NPIs such as *jemals* (‘ever’) can cause illusions, while attenuating ones do not (*so recht* ‘really’). This result was incompatible with the predictions of the cue-based retrieval and feature percolation accounts.

To summarise this section, we can say that many of the predictions of the two accounts, which join together NPI illusions and agreement attraction, have arguably not been borne out in recent investigations. The need to account for these empirical findings has led to the development of two recent theories of NPI illusions that treat the phenomenon separately from agreement attraction, the scalar approach and the active scope approach. We discuss them below.

1.3.3 The active scope and the scalar alternative approaches

Let us start with active scope theory. Mainly to deal with the issue of the ineffectiveness of sentential negation as an illusory licenser, Orth et al. (2021) proposed a new, *quantificational*, theory of how NPI illusions arise—the active scope theory. As it is well known that when it comes to quantifiers, these expressions can be interpreted in a different position than would be suggested by the surface form of the sentence, due to the phenomenon of quantifier raising. NPI illusions arise with negative quantifiers such as *no* due to them taking scope over the whole matrix clause in which the NPI is located. The account claims that quantifier raising is always attempted when the negative quantifier *no* is encountered by comprehenders. Even though this is then found to be impossible, the fleeting representation in which the quantifier scopes over the NPI is constructed. Thus, the NPI can be erroneously licensed, since it is found within the scope of a licenser. This is

³Note that Thorsen (2025) reports that in the Germanic language of Danish, sentential negation (*ikke*) gives rise to an illusion with the NPI *nogensinde* (‘ever’) in a 7-point Likert scale judgement task. However, these results ought to be treated with caution due to the small sample size employed and the post-hoc nature of the analysis.

unlike the construction with sentential negation, which does not involve attempted raising. Given that simple negation within the RC does not show the same scopal behaviour as quantifiers, it cannot be interpreted as scoping over the matrix clause and thus, it cannot cause the illusion of proper NPI licensing.

The active scope theory received further support from another effect—PPI illusions. Orth et al. (2025) tested whether PPIs (positive polarity items) such as *still*, which are anti-licensed by negation (Giannakidou, 2019; Homer, 2020; Szabolcsi, 2004), give rise to an illusion of *ungrammaticality*, that is cases where comprehenders mistakenly consider grammatical sentences as ungrammatical. Orth et al. (2025) found that when an RC that includes the negative quantifier *no* is inserted into an affirmative sentence with the PPI *still*, comprehenders came under the illusion that said sentence is ungrammatical. This effect is predicted by the active scope theory, since the quantifier *no* attempts raising and temporarily scopes over the PPI. Since PPIs are *anti*-licensed by negation, the result is the seeming ungrammaticality of the PPI. Thus, the active scope theory says that the attempted quantifier raising of *no* can either create illusions of grammaticality or ungrammaticality depending on the type of polarity item in question. Sentential negation is said to be incapable of causing either effect.

Let us now turn to the scalar alternative approach. The theory (Muller, 2022; Schwab, 2023) claims that NPI illusions arise due to comprehenders failing to suppress certain inappropriate scalar alternatives that could under other conditions license NPIs. This is based on the scalar approach to NPI licensing (Fauconnier, 1975; Crnić, 2011; Chierchia, 2013), which claims that NPIs are allowed only insofar as the assertion in which the expression is found is stronger compared to its alternatives (Crnić, 2011; Chierchia, 2013). Let us take the example of the NPI *ever*:

(11) The student hasn't ever got an A.

In (11), the alternatives are that the student has not got an A once, twice, many times and so on. Without negation, contradiction is said to follow (see Krifka, 1995; Crnić, 2011; Chierchia, 2013, and others for formal implementations of this idea). Schwab (2023) applies this theory to the illusion cases and argues that it can account for NPI illusions only arising with certain types of NPIs and also not with sentential negation as an illusory licenser. Remember that only strengthening NPIs were found to be susceptible to illusions in her study, while attenuating ones are not. Attenuating NPIs are said to require that the proposition in the assertion to be weaker compared to the alternative propositions (Schwab & Liu, 2022). She argued that this pattern of difference could be best explained by appealing to the scalar alternative theory. When it comes to the strengthening German NPI *jemals*, the situation is equivalent to the English *ever*. The RC with the negative quantifier provides the right type of alternatives that could potentially license the NPI. This is because the quantifier *no* is said to activate scalar alternatives such as *some* or *many*—the right type of scale. In the case of attenuating NPIs on the other hand, the alternatives

evoked by the RC, which includes the negative quantifier, are in fact stronger. Consequently, they do not provide the possibility for illusory licensing. This is because the alternatives are not of the right type. In the crucial case of sentential negation, illusory licensing also cannot arise, since sentential negation is said to only activate the alternative of the proposition of opposite polarity (true or false), which is again the wrong type of alternative for strengthening NPIs such as *ever*.

Let us now summarise the approaches to explaining NPI illusions postulated in the literature, their strengths and shortcomings. The cue-based retrieval theory (Vasishth et al., 2008) provides a general explanation for both NPI illusions and agreement attraction. However, it struggles with explaining (a) why only certain NPIs and NPI licensors are susceptible to illusions, (b) the different vulnerability profiles of NPI illusions and agreement attraction, and (c) the existence of the PPI illusion. Feature percolation approaches (e.g., Eberhard et al., 2005) can account for why certain NPI licensors are not effective illusory ones but arguably not why only certain NPIs are susceptible. The scalar theory of NPI illusions (Schwab, 2023; Muller, 2022) based on the formal treatments of Crnić (2011) and Chierchia (2013) is able to cover the empirical ground of both NPI and PPI illusions. The active scope theory (Orth et al., 2021; Orth, 2023; Orth et al., 2025) is likewise so far capable of explaining the data so far. Crucially for the purposes of the current study, neither of these latter theories predict sentential negation to be an effective illusory licensor. In the next section, we discuss what predictions these theories have for the two different types of expressions in Czech, namely NPIs and NCIs, with regards to illusions.

1.4 Comparing NPI and NCI illusions in Czech

Our aim in the current study is to test the above-introduced theories of NPI illusions and agreement attraction by comparing the illusory licensing susceptibility of two types of negatively dependent expressions within a single language, namely Czech. We will argue that the effectiveness of sentential negation as an illusory licensor can be used as a diagnostic for these approaches. Let us now go through the predictions.

Firstly, when it comes to the two approaches that unite NPI illusions with agreement attraction in terms of explanatory mechanisms, they both predict that sentential negation should be an equally good illusory licensors for both Czech NPIs and NCIs, regardless of the different nature of their licensing; that is with one being in the syntax and the other semantically/pragmatically dependent, as we argued. With the cue-based retrieval, this is easy to see, as the sentential negation morpheme *ne-* in Czech presumably carries the “+negative” feature and can be misretrieved. With feature percolation, sentential negation can be seen as effective as long as negation in Czech is not found too low in the syntactic structure to percolate up. According to linguistic analyses of Czech, negation is indeed found high in the structure (Zeijlstra, 2022) and thus potentially capable of influencing the negativity of the matrix clause. We will return to this point in the General discussion Section

4.

Next, let us discuss the two recent approaches of the active scope and scalar alternative theories. When it comes to Czech NPIs, these theories both predict that sentential negation should be entirely ineffective as an illusory licenser. In the case of active scope, this is because the theory claims that sentential negation does not attempt raising unlike quantifiers (Orth et al., 2021, 2025). The scalar alternative approach (Schwab, 2023; Muller, 2022) bars the possibility of illusions due to sentential negation not activating the right type of alternatives. The Czech NPI *ani* (‘not even’) is a strengthening NPI (Dočekal & Dotlačil, 2024) and thus, like the English *ever* and German *jemals* (‘ever’), requires alternatives of an ordered scalar nature (see example (4-a)).

When it comes to NCIs, neither of these theories makes an explicit claim regarding this type of expressions. With the active scope theory, we could claim that the theory predicts no illusory licensing in virtue of the attempted raising mechanism. The recent NCI illusion results of Lacina (2025) are, however, compatible with it, since in that study, negative quantifiers were found in the intervening RC, and thus could have given rise to the illusions. If this is indeed the source of the effect, the active scope theory predicts no NCI illusions with sentential negation only. However, as we argued, NCI licensing could be of an entirely different, agreement-like, nature and thus other mechanisms such as misretrieval could still cause illusions. The scalar alternative theory is entirely silent when it comes to NCIs. As we argued based on experimental semantic evidence in Section 1.1, while Czech NPIs are sensitive to pragmatics, NCI licensing is not affected by these factors. Therefore, the scalar theory does not apply and does not predict anything with regards to illusions with NCIs.

Given the earlier results from English showing the ineffectiveness of sentential negation as an illusory licenser, it thus appears that we have good reason to suspect that in Czech, simple sentential negation, while capable of licensing NPIs within its scope (Dočekal, 2014, 2017; Dočekal & Dotlačil, 2016, 2018), ought not to be an *illusory* licenser in the language. The explanation for why negation is effective in languages such as Turkish (e.g., Yanilmaz & Drury, 2018), Korean (e.g., Yun et al., 2018) and Japanese (Su & Aparicio, 2022), namely that these languages are SOV and that they tested *prospective* licensing configurations, does not apply to Czech. The language is said to have free word order (see Jasinskaja & Šimík, in press, for a recent overview of Slavic word order) with all the possible orderings of the basic arguments S, V, and O allowed by the grammar. However, these are not all equally frequent in the language with the SVO order being the most common and considered canonical (Siewierska & Uhlířová, 1998). Liu (2010) examined the head directionality of dependencies in the corpora of twenty languages and found that in Czech, only slightly more than half of phrases (54.5%) were head-initial with the rest being head-final (45.5%). What this tells us is that unlike languages such as Turkish (SOV, over 94% head-final according to Liu (2010) or 86% head-final as reported by Türk et al. (2022)), Czech can construct sentences similar to the English NPI illusion constructions where the NPI follows its licenser, i.e.,

retrospective constructions. Therefore, we expect no illusions with Czech NPIs when sentential negation is present in an intervening relative clause.

The same, however, might not be the case with NCIs. As discussed above, NCI licensing has been argued to be a type of syntactic agreement compared to the semantic/pragmatic nature of NPI licensing. As such, the cue-based retrieval model could still be applicable to NCI illusions in the same way that it deals with agreement attraction. This is because should NCI licensing happen in the way that the syntactic theories suggest (Zeijlstra, 2008), the mechanism would simply require the retrieval of the element with the negative feature. NCI illusions would then occur in the same way that agreement attraction happens—by means of the mistaken retrieval of the wrong element in the situation where there is only a partial feature match with the true controller of agreement (Logačev & Vasishth, 2016). Likewise, the feature percolation account could argue that negation percolates up and causes the representation of the matrix clause to be more negative. However, this account would struggle to explain how this could be different for NPIs and NCIs. In sum, assuming the syntactic agreement treatment of NCIs to be right, we may predict that sentential negation ought to be effective in creating illusions with NCIs.

Additionally, as Orth et al. (2021) point out in their article, the cross-linguistic coverage of the NPI illusion phenomenon has so far been sparse with no Slavic language tested so far to our knowledge. A direct comparison of the NCI and NPI illusions in a language that has both types of expressions is also missing, although it could shed light on the underlying mechanism or mechanisms that support the online licensing of the two phenomena. Parker (2025a) makes this point exactly when discussing advised future directions of the research on the processing of negation. In his view, NCIs and NPIs are also expected to differ in processing due to their semantic/pragmatic and syntactic nature respectively. In the current study, therefore, we set out to test whether the prediction that sentential negation should act as an illusory licenser for NCIs but not for NPIs holds. We conducted speeded acceptability judgement experiments with native speakers of Czech where we used sentential negation as the potential illusory licenser with the NPI *ani* ('not even', Experiment 1 and 3) and with the NCI *žádný* ('no', Experiment 2).

2 Experiments 1 and 2

In two speeded acceptability judgement experiments, we probed for the presence of grammaticality illusions in Czech with NPIs (Experiment 1) and with NCIs (Experiment 2). The aim of our study was to test whether simple sentential negation can elicit NPI and NCI illusions in the online comprehension of the language. Given the earlier results from English where sentential negation was insufficient to trigger illusions (Orth et al., 2021; Muller, 2022), we expected no effect with NPIs. However, as NCI licensing has been argued to be syntactic agreement in nature (Zeijlstra, 2004, 2008), the simple presence of a negative marker within an intervening RC ought to trigger

illusions for this type of expressions. Our hypothesis was the following:

Hypothesis (A): Sentential negation within an embedded relative clause will cause increased acceptance for unlicensed NCIs but not unlicensed NPIs within the matrix clause.

2.1 Method

2.1.1 Participants

We recruited participants from a student participant pool at Charles University, Prague, who received course credit for their participation. This was 158 people in Experiment 1 and 191 in Experiment 2. The samples did not overlap. In Experiment 1, there were 35 men, 120 women, two persons of other gender and one person who did not disclose their gender (mean age of 23.1, SD = 6.1). In Experiment 2, the demographics were the following: 151 women, 39 men and one person who declined to answer the gender question (mean age of 22.6, SD = 4.5). We only included native speakers of Czech who reported not having grown up speaking another language in addition to Czech. The resulting exclusions based on this criterion were the following for the two experiments. In Experiment 1, we excluded 15 people and in Experiment 2, it was 12 participants.

2.1.2 Materials

For each experiment, we created 32 sets of sentence quartets with each version corresponding to one of our four conditions. We manipulated whether the critical NPI *ani* ('not even') in Experiment 1 or the NCI *žádný* ('no') in Experiment 2 within the matrix clause was licensed or not. This was the LICENSING factor (LICENSED or UNLICENSED). Next, we manipulated whether negation was present in an intervening relative clause—the RC NEGATION factor (AFFIRMATIVE or NEGATED). The reader may find example items below in Tables 1 and 2.

Table 1: Experimental item example from Experiment 1 (NPIs) together with glosses according to the Leipzig glossing rules (Comrie et al., 2008).

	1	2	3	4	5	6	7	8	9	10	11
a	Minulý	měsíc	ne-vyhrál	karbaník,	který	hledá	dlouhodobou	práci,	ani	jednu	sázku.
	Last	month	NEG-won	gambler	who	looks.for	long-term	job	even	one	bet
b	*Minulý	měsíc	vyhrál	karbaník,	který	hledá	dlouhodobou	práci,	ani	jednu	sázku.
	Last	month	won	gambler	who	looks.for	long-term	job	even	one	bet
c	Minulý	měsíc	ne-vyhrál	karbaník,	který	ne-hledá	dlouhodobou	práci,	ani	jednu	sázku.
	Last	month	NEG-won	gambler	who	NEG-looks.for	long-term	job	even	one	bet
d	*Minulý	měsíc	vyhrál	karbaník,	který	ne-hledá	dlouhodobou	práci,	ani	jednu	sázku.
	Last	month	won	gambler	who	NEG-looks.for	long-term	job	even	one	bet

'Last month, the gambler, who is/isn't looking for a long-term job, didn't win even a single bet.'

The two sets of stimuli for Experiments 1 and 2 differ in what modifies the direct object of the matrix clause (*sázku* 'bet.ACC'). This is either an NPI or an NCI. In the design, conditions in (d) are crucial, since these include negation within the RC, which, according to our predictions, ought to be affective in raising comprehenders' acceptance of the unlicensed NCI in Experiment 2, but

Table 2: Experimental item example from Experiment 2 (NCIs) together with glosses according to the Leipzig glossing rules (Comrie et al., 2008).

	1	2	3	4	5	6	7	8	9	10
a	Minulý Last	měsíc month	ne-vyhrál NEG-won	karbaník, gambler	který who	hledá looks.for	dlouhodobou long-term	práci, job	žádnou no	sázku. bet
b	*Minulý Last	měsíc month	vyhrál won	karbaník, gambler	který who	hledá looks.for	dlouhodobou long-term	práci, job	žádnou no	sázku. bet
c	Minulý Last	měsíc month	ne-vyhrál NEG-won	karbaník, gambler	který who	ne-hledá NEG-looks.for	dlouhodobou long-term	práci, job	žádnou no	sázku. bet
d	*Minulý Last	měsíc month	vyhrál won	karbaník, gambler	který who	ne-hledá NEG-looks.for	dlouhodobou long-term	práci, job	žádnou no	sázku. bet
‘Last month, the gambler, who is/isn’t looking for a long-term job, won no bet.’										

not with the NPI in Experiment 1. We also constructed a set of 64 filler items that were of varying structure. Some of the filler items were grammatical while others were not. In total, the number of grammatical and ungrammatical sentences was balanced in the combined critical item–filler set.

2.1.3 Procedure

Both experiments were hosted on PCIBex (Zehr & Schwarz, 2018). Participants were given a link and could take part at their convenience within one week. They were first given instructions and then saw eight practice items. Within the experiments themselves, they saw 32 critical and 64 filler sentences presented using the RSVP method (Potter, 2018) with 350ms per word. Following the end of each sentence, their task was to judge within 3000ms whether the sentence just presented seemed natural to them or not by pressing ‘j’ for ‘yes’ and ‘f’ for ‘no’. The Latin Square design was used and therefore, no participant saw more than one version of each item. They each judged eight instances of each condition.

2.1.4 Analysis

The obtained data from both experiments were first cleaned—we excluded all trials where the participant’s response time was less than 200ms, an amount of time we deemed too short for them to be able to make a judgement on the assumption that they actually paid enough attention to the presented material. This resulted in 4.15% of data loss in Experiment 1 and 6.27% in Experiment 2. One participant in Experiment 2 was removed due to them having the mean value of RTs below 200ms. On this cleaned dataset, we fit a generalised linear mixed effects model using the *lmer* and *lmerTest* packages (Kuznetsova et al., 2017) in the programming language R (R Core Team, 2023). We used participants’ binary responses (*yes* or *no*) as the dependent variable. The model had a nested structure, as we aimed to test the effect of RC NEGATION within both the LICENSED and, crucially, the UNLICENSED conditions. This allowed us to test whether there were any differences present within the ungrammatical UNLICENSED conditions. The LICENSED and NEGATION conditions were coded as 1, the UNLICENSED and AFFIRMATIVE as -1 . In terms of the random effects structure employed, in Experiment 1, the model converged with the full random effects structure allowed by the design (Barr et al., 2013), that is random intercepts for both

participants and items and random slopes for all the fixed effects for both participants and items. In Experiment 2, we simplified the random effects structure for items to only include the random slope for the effect of LICENSING.

2.2 Results

Participants' responses following data cleaning by condition can be found in Figures 1 (Experiment 1, NPIs) and 2 (Experiment 2, NCIs) respectively.

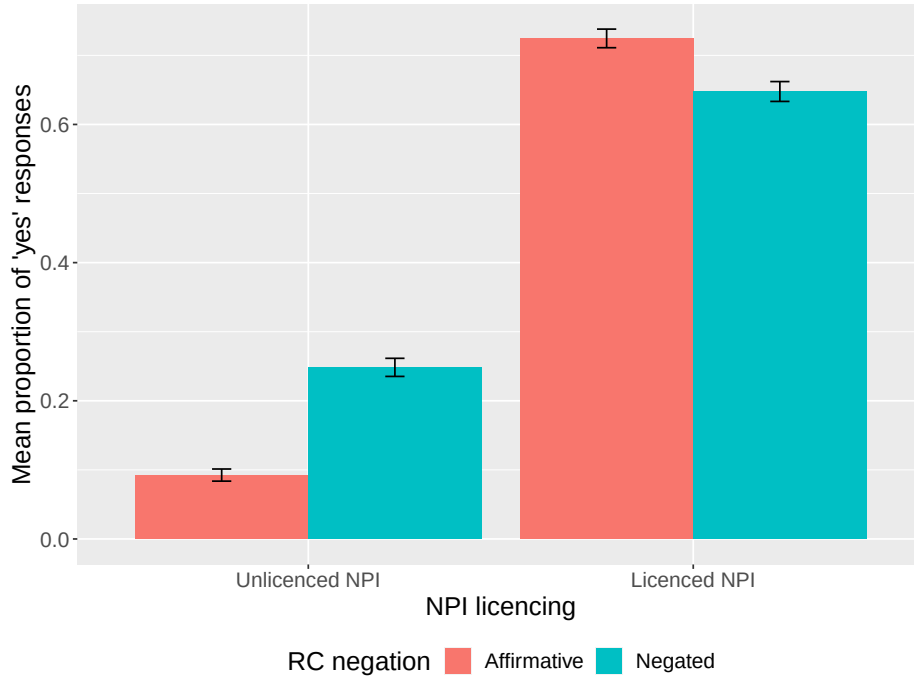


Figure 1: Mean ratings by condition from Experiment 1 (NPIs) with SEs.

The results of the nested models were the following. Let us take Experiment 1 (NPIs) first. The model revealed a significant effect of the LICENSING condition. UNLICENSED NPIs in the matrix clause caused the rate of acceptability to decrease when compared to grammatical sentences. The values obtained were the following: $\beta = 1.80260$, $SE = 0.09272$, $z = 19.442$, $p < 0.0001^{***}$. The effect of RC NEGATION was significant within the UNLICENSED conditions and also within LICENSED ones, however the direction of the effect was the opposite. In the former, the presence of negation in the RC increased acceptability, in the latter, it decreased acceptability rates. The values for the UNLICENSED conditions were: $\beta = 0.65349$, $SE = 0.08086$, $z = 8.082$, $p < 0.0001^{***}$. For the LICENSED ones, the output of the model was the following: $\beta = -0.18633$, $SE = 0.06531$, $z = -2.853$, $p = 0.00433^{**}$.

We now turn to the results of Experiment 2 (NCIs). Here, the pattern was identical to the one seen in Experiment 1 (NPIs). We found that sentences with UNLICENSED NCIs were accepted significantly less often compared to sentences with LICENSED ones ($\beta = 1.57511$, $SE = 0.08368$, $z = 18.822$, $p < 0.0001^{***}$). The RC NEGATION factor was statistically significant within both

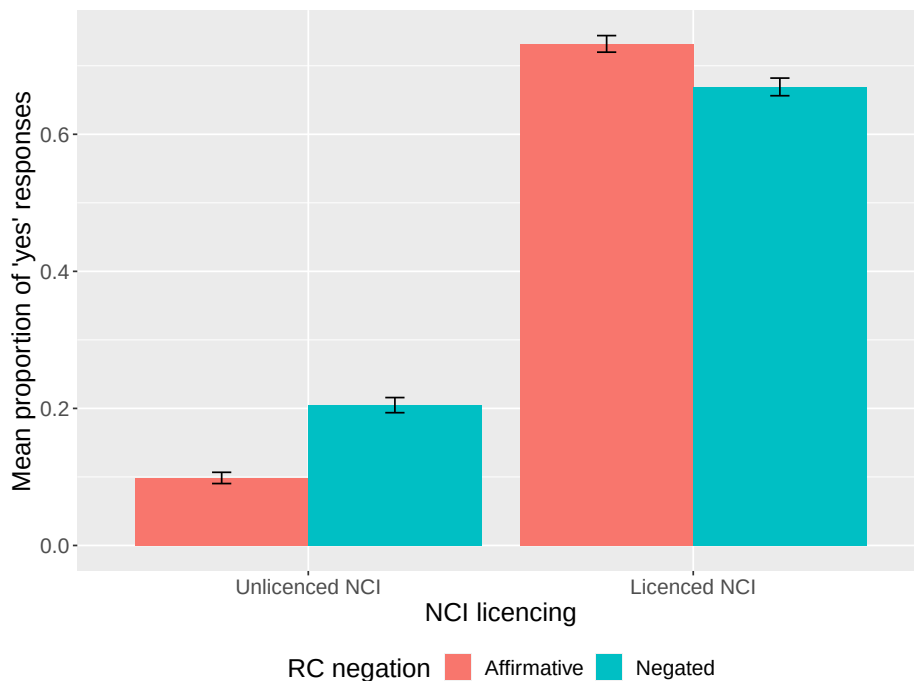


Figure 2: Mean ratings by condition from Experiment 2 (NCIs) with SEs.

the UNLICENSED and LICENSED conditions. The values for the factor within UNLICENSED sentences were: $\beta = 0.78989$, $SE = 0.07869$, $z = 10.038$, $p < 0.0001^{***}$. The values for the LICENSED conditions stood at: $\beta = -0.20701$, $SE = 0.05894$, $z = -3.512$, $p < 0.0001^{***}$.

2.3 Discussion

In Experiments 1 and 2, we asked whether sentential negation serves as an illusory licenser for Czech NPIs and NCIs respectively. We predicted that grammaticality illusions would occur only with the NCIs in Experiment 2, given that their licensing is argued to lie in the syntax (Zeijlstra, 2008); while recent findings in English (de Dios Flores et al., 2017; Orth et al., 2021; Muller, 2022) and theoretical approaches such as the active scope theory (Orth et al., 2021, 2025) and the scalar theory (Muller, 2022; Schwab, 2023) predicted that in the case of NPIs (Experiment 1), sentential negation would prove ineffective in giving rise to illusions (see Hypothesis A). We ran two speeded acceptability judgement experiments in order to test these predictions. What we found was only partly in line with our expectations—we report that in Czech, sentential negation gives rise to illusions with both NCIs and NPIs.

The results of Experiment 1, which tested the NPI *ani* (‘not even’) are consistent with the interpretation that Czech comprehenders came under a grammaticality illusion. This was because the rate of acceptance for sentences with unlicensed NPIs was higher in the condition where an intervening RC contained a negated verb than in the condition where the RC did not include negation. We saw a substantial increase in the perceived acceptability in the configuration typical for NPI illusions. We saw the opposite in the grammatical conditions where the presence of negation

in the RC led to degraded acceptability.

This pattern was repeated in Experiment 2, in which we examined the NCI *žádný* (the negative quantifier ‘no’). Again, the presence of negation in the RC caused comprehenders to come under the illusion that unlicensed NCIs in the matrix clause were acceptable. This result supports the earlier findings of [Lacina \(2025\)](#), who reported RCs with negative quantifiers and negation to be effective in causing increased acceptability for otherwise ungrammatical sentences with the same unlicensed NCI as used in the current study.

We will return to discussing the overall implications of these findings in the General discussion Section 4. We must, however, first discuss a potential alternative explanation for the results of Experiments 1 and 2 that does not treat them as illusions.

There is the possibility that our results could be explained as cases of garden-path effects (for evidence of the well-known phenomenon in Czech, see [Chromý, 2022](#); [Ceháková & Chromý, 2023](#)) rather than as NPI and NCI illusions respectively. These effects occur when a certain substring of a sentence provides an initially temporarily grammatical parse, although overall the parse is ungrammatical when considered within the context of the whole sentence. The reason we could potentially attribute our results to this effect is that the direct object phrase that concluded our sentences in both experiments could be understood as attaching to the RC rather than being a part of the matrix clause. Take the following illustrative example:

- (12) ..., který ne-hledá dlouhodobou práci, žádnou sázku a žádnou partnerku.
..., who NEG-seeks long-term job.ACC, no bet.ACC and no partner.ACC.
‘..., who is not looking for a long-term job, any bet and no partner.’

What we see in (12) is a possible continuation of the RC type that we used in our Experiment 2 (see 2 for the reference item). We see that following the direct object (*práci*, ‘job.ACC’) of the verb *nehledá* (‘NEG-seeks’), the sentence continues with the phrase *žádnou sázku* (‘no bet.ACC’), which attaches to the verb in the matrix clause rather than to the one in the RC. However, at this point, this phrase could also be a continuation of the RC as a part of a conjoined direct object with further conjuncts following it. This is what the reader can see in the example (12) above with *a žádnou partnerku* (‘and no partner.ACC’). Although this parse is ungrammatical with no third conjunct after the critical phrase with the NPI/NCI, our probing happened directly following the presentation of the sentence, which concluded with the critical phrase. Therefore, this parse could have lingered (see [Christianson et al., 2001](#); [Y. Huang & Ferreira, 2021](#); [Slattery et al., 2013](#), for evidence of this with regards to interpretations) and influenced the acceptability ratings, as in the case of the conditions with an embedded negation in the RC, this parse would have contained an NPI/NCI licenser. It must be noted, however, that recently, similarly locally coherent structures have been found not to lead to any illusions of grammaticality ([Paape et al., 2021](#)).

In order to rule this possibility out, we conducted a further NPI Experiment 3 described below

in Section 3, in which we aimed at testing sentences where the above-described issue does not arise.

3 Experiment 3

In order to circumvent the issue of the results of Experiments 1 and 2 being potentially due to parsing effects, we modified the RC in which negation can act as an illusory licenser. Specifically, the items in Experiment 3 used intransitive verbs in the RC in place of the transitives used in Experiments 1 and 2. In this way, the direct object *sázku* (‘bet.ACC’) cannot be in any way attached to the RC as in (12). Take an example of below:

- (13) *Minulý měsíc vyhrál karbaník, který ne-přespává v hotelu, ani jednu sázku.
Last month won gambler who NEG-sleeps.over in hotel even one bet
‘Last month, the gambler, who isn’t sleeping over in a hotel, won even a single bet.’

The verb *přespávat* (‘sleep over’) cannot take the noun *sázka* (‘bet’ or ‘wager’) and thus, the problematic parsing possibility is eliminated. The set up in (13) ensures that any increased acceptability in ungrammatical sentences with an illusory licenser can be considered cases of NPI illusions. This is what we predicted in case Experiments 1 and 2 are in fact illusion phenomena. Should no increased acceptability be seen in this experiment, this would be evidence in favour of parsing effects being at the root of the pattern observed.

3.1 Method

3.1.1 Participants

The experiment was conducted as a part of a psycholinguistics course at Masaryk University, Brno. Students were tasked with finding volunteer participants among their acquaintance. First, we had exclude three participants, since they indicated that they were below 18 years of age (even though the study was aimed at adults). The data of these three participants were deleted and are not included in the results file shared on OSF. 70 participants recruited for Experiment 3 remained. Overall, there were 45 women and 25 men with the mean age of 28.5 years (SD = 12.0). We excluded three other participants due to them not being native speakers of Czech.

3.1.2 Materials

We created a set of 24 items with four versions each. Students participated in the creation of the stimuli (see the Acknowledgements Section 5). The items were analogous in structure to those used in Experiment 1 (NPIs) with the NPI *ani* (‘not even’), differing in the presence or absence of the negative marker *ne-* in the matrix and relative clause respectively. The items differed from those in Experiments 1 and 2 in that the verb in the RC was intransitive (*přespávat* ‘to sleep

over’ in the example below). In this way, we ensured that the direct object phrase containing the NPI could not be plausibly attached to the RC. This eliminated the potential confound present in Experiments 1 and 2. See an example item in Table 3:

Table 3: Experimental item example from Experiment 3 (NPIs) together with glosses according to the Leipzig glossing rules (Comrie et al., 2008).

	1	2	3	4	5	6	7	8	9	10	11
a	Minulý Last	měsíc month	ne-vyhrál NEG-won	karbanfk, gambler	který who	přespává sleeps.over	v in	hotelu, hotel	ani even	jednu one	sázku. bet
b	*Minulý Last	měsíc month	vyhrál won	karbanfk, gambler	který who	přespává sleeps.over	v in	hotelu, hotel	ani even	jednu one	sázku. bet
c	Minulý Last	měsíc month	ne-vyhrál NEG-won	karbanfk, gambler	který who	ne-přespává NEG-sleeps.over	v in	hotelu, hotel	ani even	jednu one	sázku. bet
d	*Minulý Last	měsíc month	vyhrál won	karbanfk, gambler	který who	ne-přespává NEG-sleeps.over	v in	hotelu, hotel	ani even	jednu one	sázku. bet
‘Last month, the gambler, who is/isn’t sleeping over in a hotel, didn’t win even a single bet.’											

The crucial condition was, just as in Experiments 1 and 2, (d) where negation is absent in the matrix clause, but present in the relative one. This is where increased acceptability rates ought to be seen (as compared to the condition in (b)) in case Czech comprehenders are sensitive to NPI illusions.

Additionally, 48 filler items were created. The expected answer as to their acceptability was ‘no’ to one half and ‘yes’ to the other. Since the experimental items were also split in half according to grammaticality, the number of correct yes and no responses was balanced.

3.1.3 Procedure

The procedure was the same as in Experiments 1 and 2 with the following difference: Participants were not a part of a participant pool and rather than receiving a link from the experimenters directly, they were first approached by students of Masaryk University from whom they obtained the link.

3.1.4 Analysis

The analysis was similar to that of Experiments 1 and 2 with the only difference being the method of participant exclusion. In this experiment, we used filler items to exclude participants who did not pay enough attention to the experiment. The threshold of at least 80% of correct responses to filler items was set. Nine participants did not meet this threshold and were subsequently excluded from the dataset.

3.2 Results

A graphical summary of the acceptability judgement data across conditions can be viewed in Figure 3 below.

The results of the nested model we ran revealed the following patterns. Firstly, there was a significant effect of the LICENSING condition ($\beta = 4.3273$, $SE = 0.4357$, $z = 9.933$, $p < 0.0001^{***}$)

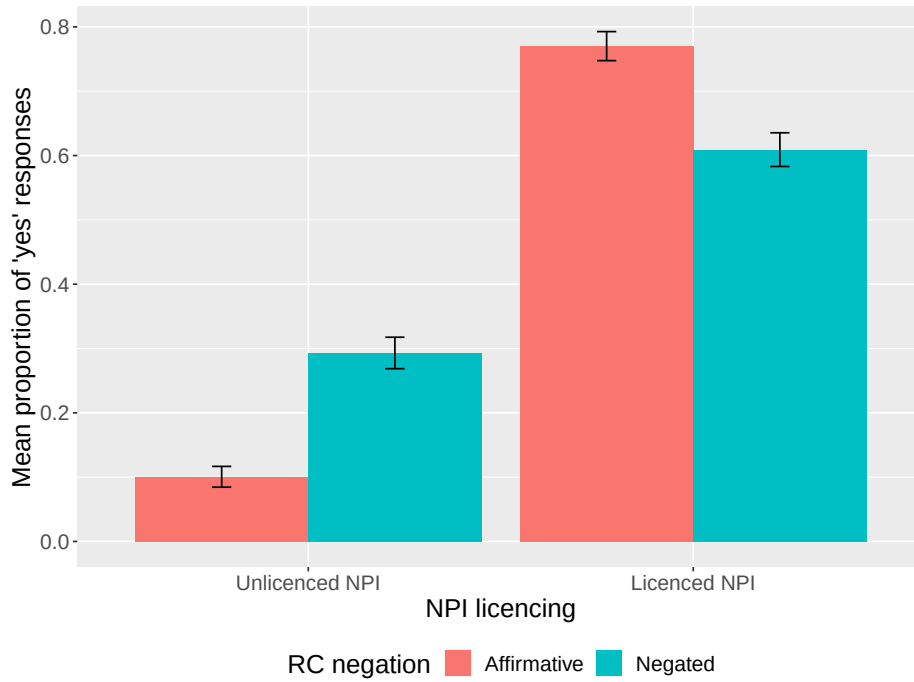


Figure 3: Mean ratings by condition from Experiment 3 (NPIs) with SEs.

with UNLICENSED NPIs being rejected at higher rates compared to LICENSED ones. Within the LICENSED condition, the factor of RC NEGATION had a significant negative effect ($\beta = -1.1481$, $SE = 0.44243$, $z = -2.595$, $p = 0.0097^{**}$) showing a decrease in acceptability rates when the RC contained negation. Crucially, the presence of negation was found to be significant in the positive direction within the UNLICENSED condition ($\beta = 1.4872$, $SE = 0.3372$, $z = 4.411$, $p < 0.0001^{***}$).

3.3 Discussion

In the Discussion Section 2.3 to Experiments 1 and 2, we laid out the worry that the increased acceptability rates in the critical conditions could in fact be reflective of incorrect, but temporarily plausibly parses due to comprehenders possibly attaching the NPI/NCI-containing direct object to the relative rather than the matrix clause. In Experiment 3, we replaced the transitive verbs in the RC with intransitive ones to avoid this confound.

The results of the experiment clearly speak in favour of an NPI illusion, as we saw a substantial increase in the acceptability rate of the unlicensed NPI *ani* ('not even') when negation was present in the RC. This was even though the direct object with the NPI could not be attached to the RC. Thus, we can dismiss this potential explanation and as a result, advance the interpretation that in all three experiments, Czech comprehenders showed vulnerability to grammatical illusions.

4 General discussion

4.1 Results summary

In the current study, we aimed at testing whether Czech comprehenders come under grammaticality illusions when exposed to unlicensed NPIs and NCIs that nevertheless appear in the company of a preceding relative clause (RC), which includes the possible illusory licenser in the form of the sentential negation marker *ne-* attached to the verb. Based on earlier findings that suggested sentential negation to be ineffective as an illusory licenser for NPIs in English (de Dios Flores et al., 2017; Orth et al., 2021; Muller, 2022), we predicted that in the case of Czech NPIs, no illusions would likewise be observed. On the other hand, we predicted that in the case of NCIs (n-words, neg-words), which are claimed to be licensed in the syntax as agreement (Zeijlstra, 2004, 2008, 2022), illusions would occur with sentential negation as the sole illusory licenser. This was summarised in our Hypothesis A in Section 1.

Contrary to our predictions, both NCIs (Experiment 2) and NPIs (Experiments 1 and 3) showed evidence of comprehenders coming under strong grammaticality illusions. In Experiments 1 and 2, including an RC with a negated transitive verb caused comprehenders to accept unlicensed NPIs or NCIs respectively. The data showing that NCIs (or neg-words) are susceptible to an illusion effect corroborates the earlier findings of Lacina (2025), who found that RC containing negated verbs and negative quantifiers cause the illusion for the NCI *žádný* ('no'). The current experiment constitutes, to our knowledge, the first reporting of NPI illusions in a Slavic language. Finally, Experiment 3 was designed to rule out a non-illusory parsing explanation of the results of the first two experiments. In this experiment, RCs with negated intransitives caused the same effect for NPIs as negated transitives did. Thus, we can dismiss the alternative explanation and assert that our data support the existence of NPI and NCI illusions caused by sentential negation in Czech comprehenders.

4.2 Implications for processing theories

We now turn to what the current data mean for the various theories of online NPI licensing that have been advanced in the literature. Recall the four theories that we reviewed in the Introduction Sections 1.3.1 and 1.3.3—the cue-based retrieval theory (Vasishth et al., 2008), the feature-percolation account (Eberhard et al., 2005), the active scope theory (Orth et al., 2021), and the scalar alternatives theory (Muller, 2022; Schwab, 2023). While the first two approaches either predicted sentential negation to serve as an illusory licenser or at least be made compatible with its effectiveness, the latter two predicted no illusions to be caused by sentential negation. In the active scope theory, this is because sentential negation cannot attempt raising out of the RC to temporarily scope over the NPI (Orth et al., 2021, 2025). The scalar theory (Muller, 2022;

Schwab, 2023) predicts no effects for sentential negation, since it activates the wrong type of alternatives. The results of our Experiments 1 and 3 with Czech NPIs are, therefore, compatible with the cue-based and feature-percolation accounts and incompatible with the active scope and scalar alternatives theories. We will discuss the implications of this finding below and argue in favour of the feature percolation account.

On the other hand, the NCIs tested in Experiment 2 behaved as we predicted—sentential negation was an effective illusory licenser. This is in line with the story provided by the cue-based and feature percolation theories under the assumption that NCIs are licensed in the syntax analogously to the way that morphosyntactic agreement functions (Zeijlstra, 2008). Had NCIs behaved as NPIs would behave according to the predictions of the active scope theory, we ought not to have seen any effect of sentential negation on the acceptability of unlicensed NCIs. The scalar approach, we argued, is silent on the matter of NCIs, as it specifically highlights the semantic/pragmatic nature of NPI licensing, which is absent with NCIs. As with the NPI Experiments 1 and 3, the results of the Experiment 2 support the cue-based retrieval and feature percolation approaches in their explanations of the NCI (or negative concord) illusion.

Let us now discuss whether the current study favours one of the two theories that are data are consistent with, namely cue-based retrieval and feature percolation. We argue that it is the latter that the current results support more strongly when we consider the cross-linguistic contrast in the effectiveness of sentential negation as an illusory licenser observed between English and our current Czech data.

Let us start with the argument in favour of cue-based retrieval. The support for the theory is compounded by the fact that we observed a *decrease* in acceptance rates for those sentences that contained licensed NCIs yet also sentential negation in the RC. This is because the cue-based retrieval theory predicts there to be *illusions of ungrammaticality* in cases of cue overload, i.e., several items in memory that have the matching features required by the probe of the element establishing a dependency (Logačev & Vasishth, 2016). However, we believe that this decrease is more likely the result of the pragmatic infelicity of sentences where both the matrix clause and the relative clause are negated. In general, negated sentences are less felicitous and more difficult to process when they occur without supporting context (see Kaup & Dudschig, 2020; Parker, 2025a, for overviews), as was the case in our study.

We now turn to the feature percolation approach. As has been established by de Dios Flores et al. (2017), Orth et al. (2021), and Muller (2022), in English, simple sentential negation is not an illusory licenser. In English, sentential negation sits low in the syntactic structure as opposed to negative quantifiers that are higher in the structure of the typically studied NPI illusion constructions (see (5)). The feature percolation account predicts that the higher in the structure the illusory licenser is placed, the higher the probability of illusions occurring, as argued by Orth et al. (2021). We argue, based on independent evidence, that in Slavic languages, including Czech,

sentential negation is placed high in the structure (Zeijlstra, 2022) and can thus percolate up where English negation cannot.

The main piece of evidence is that Czech allows for the licensing of NPIs in the subject position, which is ungrammatical in English. Consider the two following examples, one from Czech and the other from English:

- (14) Ani jeden student ne-dostal za A.
 Not.even one student NEG-got for A.
 ‘Not even a single student got an A.’

- (15) *Any student didn’t get an A.

In order for the Czech NPI *ani* (‘not even’) to be licensed in (14), negation must be higher in the syntactic structure than the NPI. On the other hand, English negation is not higher than the subject, hence the NPI in that position is unlicensed (see (15)). The way this is formalised is by postulating a NegP projection in languages like Czech that carries a silent negation operator. We discuss this in detail below.

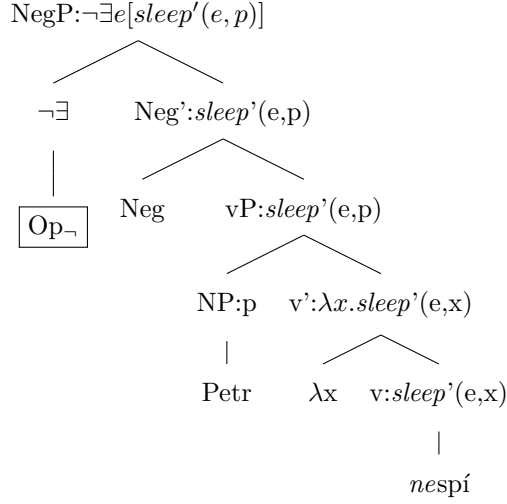
Readers are referred to the schematic syntactic trees in (17) and (18), which showcase the difference in the placement of negation in Czech and English. Both trees illustrate the structure of simple sentences with intransitive verbs such as (16) below. The trees follow Zeijlstra (2004) in treating sentential negation as a NegP projection above the vP. According to Zeijlstra (2022), languages with negative concord have the NegP projection, unlike languages without negative concord such as (Standard) English⁴. In strict negative concord languages like Czech, the NegP projection is high in the structure and hosts the semantic negation operator (Op_{-}) responsible for the classical logical semantics of negation. In English, there is no NegP projection, and the negation marker *not* is merged low in the structure within the vP. The English negation marker *not* bears the semantic negation component. To summarise, in Czech, the sentential negation marker (verbal prefix *ne-*) does not actually carry semantic negation but only agrees with the semantic negation operator merged high in the structure. The negation operator Op_{-} is the actual licenser of negative dependencies such as NPIs and NCIs and is structurally high in Czech. By contrast, the English negation marker *not* is the actual licenser of NPIs and is structurally low in the vP. The placement of negation in Czech and English respectively is highlighted by means of a rectangle encompassing the relevant node in the examples below:⁵

- (16) Petr ne-spí.
 Petr NEG-sleeps
 ‘Petr does not sleep.’

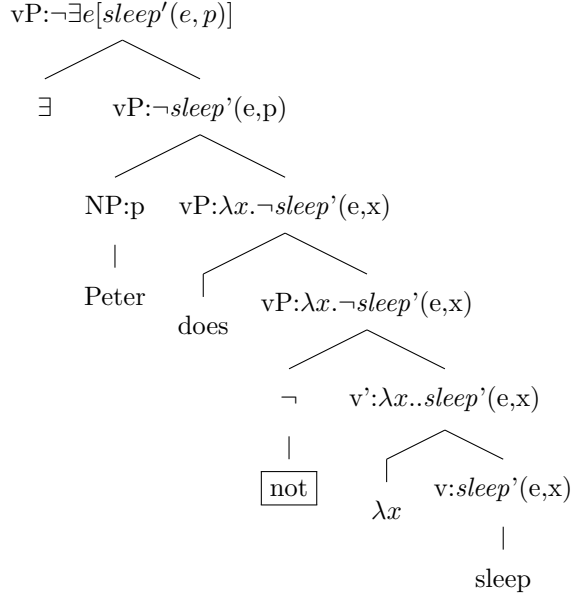
⁴But see Blanchette (2013) for a view that sees all variants of English as having negative concord but with only some (e.g., African American English) realising it.

⁵We follow Zeijlstra (2022) in treating negation as scoping over the existential closure of the event variable *e* in trees (17) and (18). However, nothing hinges on this assumption for the current discussion, and in the following argumentation, we will omit the event variable for simplicity.

(17)



(18)



Assuming that these structures are correct for Czech and English respectively, we can link them to the presence of illusions in one language and their absence the other. Recall that the feature percolation account, as argued by [Orth et al. \(2021\)](#), can make use of the embedding depth of negation to make predictions for what structures should give rise to illusions and which should not. The lower in the structure negation is found, the less likely it is predicted to percolate up and cause illusory licensing. In (17), we see that NegP, which actually hosts the interpreted negation, is the highest projection displayed, even above the subject. It is thus likely to percolate up. On the other hand, notice that in the English sentence (18), negation is interpreted low in the structure, as there is no NegP. Negation is actually interpreted on *not*, which sits below the subject. Being this low, the English negation *not* is therefore unlikely to exert any influence outside of its clause. Thus, the effectiveness of sentential negation in one language, Czech, and its ineffectiveness in another, namely English, can be explained within the feature percolation account as a consequence of the difference in the structural placement of negation.

We can marshal further, cross-linguistic, support for this claim. Turkish, another language where sentential negation causes NPI illusions (Yanilmaz & Drury, 2018; Bakay & Oğuz, 2024), is also argued to have sentential negation placed high in the syntactic structure (Kamali & Zeijlstra, 2024). Namely, in Czech as well as in Turkish, sentential negation is able to license NPIs in the subject position (Dočekal, 2017; Kamali & Zeijlstra, 2024), while in English, this is not the case. Turkish is also a negative concord language (Jeretič, 2023). Furthermore, in both Korean and Japanese, the other two languages where NPI illusions arise due to sentential negation (Yun et al., 2018; Su & Aparicio, 2022), it is also possible to license NPIs in the subject position (see Błaszczak, 2001; Shimoyama, 2011; Kim, 2001, for Japanese and Korean respectively). It is thus plausible to suggest that what was previously interpreted as a consequence of the difference between retrospective and prospective NPI licensing, could in fact be due to the differences in the placement of negation and the presence or absence of the NegP projection. We return to this point below when proposing further research directions in Section 4.4

On the other hand, the other theory we claimed to be consistent with our Czech results, cue-based retrieval, struggles to explain this cross-linguistic difference. This is because it does not operate with the syntactic placement of the element carrying the “+negative” feature during the moment of retrieval when NPI illusions are said to arise according to the theory. It does not matter whether a language has the NegP projection above the subject or not, both languages make it so that negation contributed by the sentential negation in the RC is encoded, stored and finally misretrieved when the NPI/NCI is encountered. This approach thus predicts sentential negation to be capable of eliciting illusions in both English and Czech.

Therefore, we conclude that our current results in conjunction with the previous findings from English are best explained by the feature percolation approach. This account alone can explain (a) why sentential negation is ineffective in English, (b) effective in Czech for NPIs, and it can (c) account for both the NPI and NCI illusion by appealing to the independently postulated existence of the high NegP projection that exists in languages with strict negative concord.

4.3 Alternative explanations

There are, however, some potential caveats and other explanations that we discuss below. We saw exactly the same pattern of results with both NPIs and NCIs. This pattern was more in line with our predictions for NCIs than with those for NPIs. One way to accommodate the unpredicted NPI results would be to suggest that these expressions in fact behave like NCIs in Czech and are supported by the same mechanisms as these expressions when processed online. This hypothesis is supported by the fact that in Czech, NCIs and NPIs generally behave similarly. While we have argued based on recent experimental evidence in the Introduction Section 1 that the two categories are distinct, the NPI we used in the current study still functions very similarly to NCIs.

Ani (‘not even’) is a *strong* NPI as opposed to the weak NPIs that were studied in English (*ever*, *any*). Strong NPIs have more stringent licensing conditions compared to weak NPIs—in Ladusaw’s (1992) theory, they require anti-additive contexts such as negation rather than only downward-entailing ones. Thus, strong NPIs are not licensed in questions or antecedents of conditionals where weak NPIs are licensed. It could thus be possible to maintain the active scope and scalar alternatives theories by postulating that they do not apply to the expression *ani* (‘not even’) in Czech, as this is not a weak NPI such as the English *ever*, yet a strong NPI that differs from the previously studied cases in other languages.

In their reliance on negation, therefore, strong NPIs are close to NCIs. Dočekal & Dotlačil (2024) experimentally tested the Czech NCI *žádný* (‘no’) and NPI *ani* (‘not even’) against each other in terms of licensing conditions, especially with regards to their acceptability in neg-raising contexts (see the Introduction Section 1.1). The research shows that, from an empirical point of view, the two classes are maybe not fully independent. On the one hand, there are clear differences in their licensing profiles: NPIs such as *ani* are sensitive to likelihood and to the availability of neg-raising environments, while NCIs like *žádný* obey stricter locality constraints and are not subject to the same probabilistic restrictions. On the other hand, there are also similarities: both types pattern with indefinites, both require the presence of an appropriate licensing environment, and both can appear in a largely overlapping set of morphosyntactic contexts once their licensors are present.

A particularly relevant result of this work is that it found robust inter-speaker variation in terms of how individual items are grammatically represented. For some speakers, expressions such as *ani* (‘not even’) behave like canonical strong NPIs, with sharply defined semantic-pragmatic licensing conditions; for others, the same expression patterns much more like an NCI, subject to stricter syntactic licensing and relatively insensitive to likelihood manipulations. Put differently, these data uncovered a gradient space between “strong NPI” and “NCI” analyses at the level of individual lexical items and speakers, rather than a clean categorical split. If at least some of our Czech comprehenders treated the NPI *ani* (‘not even’) to be in the same category as the NCI *žádný* (‘no’), these participants could have driven the illusion effect for the (not-quite) NPI.

However, we caution against this explanation. We reason that had this been the case, the predicted illusion effect would have been smaller for *ani* (‘not even’) compared to *žádný* (‘no’), since some participants would have still treated the *ani* (‘not even’) to be an NPI and thus invulnerable to illusions from sentential negation. This was not the case, as in a post-hoc analysis comparing the strength of the illusion between Experiments 1 and 2 where we analysed the ungrammatical subset of the data, we found that the illusion with NPIs in Experiment 1 was in fact *stronger* than the one with NCIs in Experiment 2.⁶

⁶The critical interaction term of the model between the factors of RC NEGATION and EXPERIMENT was significant ($\beta = -0.098$, $SE = 0.05$, $z = -1.99$, $p = 0.0047^*$). We omit reporting the full model here for the sake of brevity. Readers can consult the analysis script provided on the OSF page associated with this research.

If one potential alternative is that out NPIs are closer to NCIs, another way would be to consider the opposite, namely that NCIs should be treated as a subtype of NPIs. So far, we have been assuming Zeijlstra’s (2022) syntactic analysis of negative concord. However, the theoretical literature also provides analyses that treat NCIs semantically; see Ovalle & Guerzoni (2004); Kuhn (2019, 2022), among others. The semantic and syntactic approaches share some common ground in that they both agree that NCIs are treated as existentially quantified expressions at the level of truth-conditions. The negative force, which in the syntactic theory is carried by the covert Op (the bearer of the classical logical semantics of $\neg$), is in the semantic theory reformulated as a presupposition of empty reference (Ovalle & Guerzoni, 2004)⁷. The key point is that there is a contradiction between the truth-conditional existential meaning of NCIs and their non-at-issue negative meaning in the structure. This contradiction is resolved via syntactic movement above the licenser (prototypically a negated verb) in the semantic theory. The movement is proposed to be quantifier raising (QR) of the NCI by Kuhn (2022), which ensures that NCIs obey locality conditions on licensing.

Turning to NPIs, their truth-conditional meaning is likewise existential (Barker, 2018), but following the pragmatic theories of NPI licensing (Krifka, 1995; Crnić, 2011), their licensing involves a probability/entailment non-at-issue meaning component. The contradiction between the existential meaning and the non-at-issue unlikelihood presupposition is resolved via high scope of the presupposition through association with a covert *even* operator (Crnić, 2011). Again, we see parallels with the NCI case in that the resolution of the contradiction between the two meanings is achieved via split scope, syntactically for NCIs and pragmatically for NPIs. We could then tentatively suggest that both NCI and strong NPI illusions in Czech are due to a semantic mechanism whereby speakers attempt to interpret the non-at-issue meaning component higher in the structure and mistakenly associate it with the negation in the RC.

One advantage of taking the semantic route with NCIs that it could help explain the puzzling difference in illusion strength between the NCI illusion and agreement attraction. It could capture the fact that NPI/NCI illusions differ from the very weak (or completely absent) agreement-attraction illusions in Czech (Lacina & Chromý, 2022; Chromý et al., 2023a; Chromý et al., 2023b; Lacina et al., 2025). We could argue that it is surprising that Czech comprehenders show strong illusions in negative concord, which under Zeijlstra’s (2022) view, is one type of agreement, yet show minimal or no illusions with morphosyntactic agreement in number and gender. In the speeded acceptability judgement study of Lacina & Dotlačil (2024), which is the closest in terms of design to the current investigation, the researchers found strong evidence of a number agreement attraction illusion with case-syncretic attractors. Yet, this effect was very small. Comparing their data to those obtained for English by Wagers et al. (2009), they reported an increase in the chance

⁷In the dynamic reformulation of the idea (Kuhn, 2019, 2022), this is seen as a test on the cardinality of discourse referents.

of acceptance of only 4% compared to the 45% in the English study. These results contrast with the those reported here concerning both the NPI and NCI illusions. We found that the rates of acceptance in the illusory conditions were 24.8% in Experiment 1, 20.5% in Experiment 2 and 29.5% in Experiment 3, all substantially higher compared to the approximately 12% observed for the critical condition in an agreement attraction configuration (Lacina & Dotlačil, 2024).⁸ On the other hand, our acceptance rates for the illusory configuration are in line with what has standardly been seen in the literature (see Muller & Phillips, 2020). The account sketched here, however, does not explain how NPI illusions arise in English with negative quantifiers and fail with sentential negation. For now, we mark it here as a possible avenue to be explored in future work. Additionally, further experimental testing would be needed to explore the differing predictions of the syntactic and semantic theories of NCI licensing in online processing. We suggest one way in the following Section 4.4 together with other questions opened by this current research.

4.4 Future directions

In terms of the broader research picture, the current results open avenues for future experiments in the domain of NPI and NCI illusions. As argued above, our results give us reason to suspect that in Czech, strong NPIs behave the same way as NCIs in terms of illusions and consequently, that their online comprehension could be underpinned by either mechanisms shared with syntactic agreement or on the other hand, their licensing could be semantic.

An important test for this dilemma would be to examine whether the distance manipulation, which has been found to affect NPI illusions and agreement attraction differently (Parker & Phillips, 2016; Muller, 2022), also has the same effect in the case of Czech NPIs and NCIs. Recall that these studies found that while NPI illusions disappear when distance is increased, attraction effects are maintained. Should either or neither of the two be found to be unaffected by increasing the distance between the edge of the RC and the NPI/NCI in question, this would be evidence in favour of its processing being syntactically driven. On the other hand, if increasing the distance turns out to be effective, this could suggest that the illusions are caused by semantic factors having to do with the scopal boundary of the NPI/NCI licensing environment of the RC. Another test, specifically of the semantic treatment of NCIs briefly sketched above, is to make use of syntactic islands. One prediction is the following: if the NCI expression were placed inside an island, the QR mechanism, which per Kuhn (2022) ensures locality, would block the illusion effect. Arguably, neither the cue-based retrieval account or the feature percolation account would predict island-hood to matter.

The free word order of Czech discussed briefly in Section 1.4 also provides us with the possibility

⁸The variability in acceptance rates observed here (approx. 20–30%) is fully expected under recent competition-based models of sentence processing, such as the log-normal race model proposed by Paape (2024). Under this framework, the illusory ‘acceptable’ judgement arises when a good-enough processing stream—driven by superficial feature matches like the ones we propose for Czech negation—wins the race against faithful grammatical parsing or rational inference mechanisms. Our speeded task likely biases the competition in favour of this faster, heuristic-driven stream.

to directly contrast the effects of retrospective NPI/NCI licensing employed in the current study with the prospective configuration, where the NPI precedes its licensor, as seen in the studies of [Yanilmaz & Drury \(2018\)](#) on Turkish, of [Yun et al. \(2018\)](#) on Korean, and of [Su & Aparicio \(2022\)](#) on Japanese. While not strictly SOV, Czech allows this word order and encountering an NPI/NCI early in the sentence in the direct object or subject might create a strong prediction for a licensor further down the sentence as much as it does in languages such as Turkish and Korean. On the other hand, however, Czech comprehenders may also employ such predictive strategies less strongly compared to Turkish or Korean speakers, since such word orders are less frequent and non-canonical.

Following up on the discussion in [Section 4.2](#), which highlighted the connection between the effectiveness of sentential negation as an illusory licensor and the availability of NPIs in the subject position in the languages of Czech, Turkish, Korean and Japanese, we suggest the following cross-linguistic prediction: Languages which allow for NPIs in the subject position place negation high in the syntactic structure; consequently, the speakers of such languages should be vulnerable to illusory licensing of NPIs from sentential negation. On the other hand, languages without NPIs in the subject position should show immunity to the same illusory licensor.

In sum, languages such as Czech provide a yet under-explored field where the processing of negative dependencies via the systematic susceptibility of comprehenders to illusions may be fruitfully studied. This is due to the wealth of NCI and NPI expressions that are differently licensed compared to previously studied languages as well as features such as free word order that would allow researchers to test certain contrasts within a single language that have so far only been examined cross-linguistically.

5 Conclusion

In the current research, we tested whether sentential negation in an RC could serve as an illusory licensor for otherwise unlicensed NPIs and NCIs in the matrix clause in Czech. Contrary to our predictions, which expected an effect with NCIs and not NPIs, we found that negation was effective in giving rise to grammaticality illusions for both types of expressions. In the case of NPIs, this conflicts with several theories of online NPI licensing, namely those advanced by [Orth et al. \(2021\)](#)—the active scope theory—on the one hand and [Muller \(2022\)](#) and [Schwab \(2023\)](#)—the scalar alternatives theory—on the other. This is because both of these theories predicted no illusions. However, the results of both our NPI and NCI experiments are in line with two alternative approaches, namely the cue-based retrieval theory ([Vasishth et al., 2008](#)) and the feature percolation account ([Eberhard et al., 2005](#)). We argued that the latter theory ought to be adopted as it can account for the different effectiveness of sentential negation as an illusory licensor in Czech, where it does cause illusions, as opposed to English, where it does not. This is because

these languages differ in the placement of the negative operator within the syntactic structure—in Czech, it is high in the structure, while in English, it is low (Zeijlstra, 2022). Therefore, the current study provides evidence in favour of the feature percolation account of NPI illusions and highlights how differences in the underlying structure of languages can impact online comprehension.

Data availability

All of our stimuli, the anonymised data as well as the scripts used for statistical analyses are freely available for downloading on the website of the Open Science Framework here:

https://osf.io/swrkq/overview?view_only=7ae66ff662be4b2696d975196c24a192.

Acknowledgements

We would like to thank first and foremost to the participants who took our experiments. We thank Jan Chromý for giving us access to the Charles University participant pool. Next, we thank the students of the Spring Semester 2025 ‘CJJ62 Psycholingvistika a experimentální zkoumání jazyka’ course at the Faculty of Arts, Masaryk University. They participated in the stimuli creation and participant recruitment in Experiment 3. We are grateful for the feedback this research received at the AMLaP conference in September 2025 in Prague. Finally, the first author would like to thank the Psycholinguistics Lab Meeting attendees at the University of Oxford in Trinity Term 2024, to Matthew Husband who invited me to present there and in particular to Colin Phillips, who inspired this work through his seminar on linguistic illusions in that year’s Hilary Term. This work was supported by the European Regional Development Fund project “A lifetime with language: the nature and ontogeny of linguistic communication (LangInLife)” (reg. no.: CZ.02.01.01/00/23_025/0008726).

References

- Avetisyan, S., Lago, S., & Vasishth, S. (2020). Does case marking affect agreement attraction in comprehension? *Journal of Memory and Language*, 112, 104087. doi: 10.1016/j.jml.2020.104087
- Badecker, W., & Straub, K. (2002). The processing role of structural constraints on interpretation of pronouns and anaphors. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 28(4), 748.
- Bader, M., Meng, M., & Bayer, J. (2000). Case and reanalysis. *Journal of Psycholinguistic Research*, 29, 37–52.

- Bakay, Ö., & Oğuz, M. (2024). Investigating syntactic effects in np_i illusions in Turkish. In *Proceedings of the Workshop on Turkic and Languages in Contact with Turkic* (Vol. 9, pp. 1–15).
- Barker, C. (2018). Negative polarity as scope marking. *Linguistics and philosophy*, 41(5), 483–510.
- Barr, D. J., Levy, R., Scheepers, C., & Tily, H. J. (2013). Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of memory and language*, 68(3), 255–278.
- Blanchette, F. (2013). Negative concord in English. *Linguistic Variation*, 13(1), 1–47.
- Błaszczak, J. (2001). *Investigation into the interaction between the indefinites and negation. vol. 51: Studia grammatica*. Berlin: Akademie Verlag.
- Bock, K., & Miller, C. A. (1991). Broken agreement. *Cognitive psychology*, 23(1), 45–93. doi: 10.1016/0010-0285(91)90003-7
- Ceháková, M., & Chromý, J. (2023). Garden-path sentences and the diversity of their (mis) representations. *Plos one*, 18(7), e0288817.
- Chierchia, G. (2013). *Logic in grammar: Polarity, free choice, and intervention*. Oxford: OUP.
- Christianson, K., Hollingworth, A., Halliwell, J. F., & Ferreira, F. (2001). Thematic roles assigned along the garden path linger. *Cognitive psychology*, 42(4), 368–407.
- Chromý, J. (2022). When readers fail to form a coherent representation of garden-path sentences. *Quarterly Journal of Experimental Psychology*, 75(1), 169–190.
- Chromý, J., Lacina, R., & Dotlačil, J. (2023a). Number Agreement Attraction in Czech Comprehension: Negligible Facilitation Effects. *Open Mind: Discoveries in Cognitive Science*. doi: <https://doi.org/10.1162/opmi.a.00107>
- Chromý, J., Brand, J. L., Laurinavichyute, A., & Lacina, R. (2023b). Number agreement attraction in Czech and English comprehension: A direct experimental comparison. *Glossa Psycholinguistics*, 2(1).
- Comrie, B., Haspelmath, M., & Bickel, B. (2008). *The Leipzig Glossing Rules: Conventions for interlinear morpheme-by-morpheme glosses*. Retrieved from <https://www.eva.mpg.de/lingua/pdf/Glossing-Rules.pdf>
- Crnič, L. (2011). *Getting even* (Unpublished doctoral dissertation). Massachusetts Institute of Technology, Cambridge, MA.
- de Dios Flores, I., Muller, H., & Phillips, C. (2017). Negative polarity illusions: Licensors that don't cause illusions, and blockers that do. In *The 30th Annual CUNY Human Sentence Processing Conference*.

- Dočekal, M. (2014). Intervention effects: negation, universal quantifiers, and czech. *Complex Visibles Out There. Olomouc: Palacký University*, 369–387.
- Dočekal, M. (2017). N-words and NPIs: Between syntax, semantics, and experiments. In *Advances in formal Slavic linguistics* (p. 15).
- Dočekal, M., & Dotlačil, J. (2016). Experimental evidence for neg-raising in slavic. *Linguistica*, 56(1), 93–109.
- Dočekal, M., & Dotlačil, J. (2018). When is not-believing believing that not? In *Proceedings of sinn und bedeutung* (Vol. 21, pp. 387–398).
- Dočekal, M., & Dotlačil, J. (2024). Neg-words and npi’s in czech. In G. Dalmi, J. Witkoś, & P. Ceglowski (Eds.), *Strict negative concord in slavic and finno-ugric: Licensing, structure and interpretation* (Vol. 148, pp. 111–139). Walter de Gruyter GmbH & Co KG.
- Drenhaus, H., Saddy, D., & Frisch, S. (2005). Processing negative polarity items: When negation comes through the backdoor. *Linguistic evidence: Empirical, theoretical, and computational perspectives*, 145–165.
- Eberhard, K. M., Cutting, J. C., & Bock, K. (2005). Making syntax of sense: number agreement in sentence production. *Psychological review*, 112(3), 531–559. doi: 10.1037/0033-295X.112.3.531
- Engelmann, F., Jäger, L. A., & Vasishth, S. (2019). The effect of prominence and cue association on retrieval processes: A computational account. *Cognitive Science*, 43(12), e12800. doi: 10.1111/cogs.12800
- Fauconnier, G. (1975). Pragmatic scales and logical structure. *Linguistic Inquiry*, 6(3), 353–375.
- Franck, J., Colonna, S., & Rizzi, L. (2015). Task-dependency and structure-dependency in number interference effects in sentence comprehension. *Frontiers in psychology*, 6, 349. doi: 10.3389/fpsyg.2015.00349
- Giannakidou, A. (2000). Negative... concord? *Natural Language & Linguistic Theory*, 18(3), 457–523.
- Giannakidou, A. (2019). 3. negative and positive polarity items. In P. Portner, C. Maienborn, & K. von Stechow (Eds.), *Semantics - sentence and information structure* (pp. 69–134). Berlin, Boston: De Gruyter Mouton. Retrieved 2025-10-29, from <https://doi.org/10.1515/9783110589863-003> doi: doi:10.1515/9783110589863-003
- Giannakidou, A., & Zeijlstra, H. (2017). The landscape of negative dependencies: negative concord and n-words. In *The wiley blackwell companion to syntax* (pp. 1–38). Wiley Blackwell.

- Gibson, E., & Thomas, J. (1999). Memory limitations and structural forgetting: The perception of complex ungrammatical sentences as grammatical. *Language and Cognitive Processes*, 14(3), 225–248.
- Gonzalez Alonso, J., Cunnings, I., Fujita, H., Miller, D., & Rothman, J. (2021). Gender attraction in sentence comprehension. *Glossa: a journal of general linguistics*, 6(1).
- Hammerly, C., Staub, A., & Dillon, B. (2019). The grammaticality asymmetry in agreement attraction reflects response bias: Experimental and modeling evidence. *Cognitive psychology*, 110, 70–104. doi: 10.1016/j.cogpsych.2019.01.001
- Häussler, J., & Bader, M. (2015). An interference account of the missing-vp effect. *Frontiers in Psychology*, 6, 766.
- Homer, V. (2020). Domains of polarity items. *Journal of Semantics*, 38(1), 1-48. Retrieved from <https://doi.org/10.1093/jos/ffaa006> doi: 10.1093/jos/ffaa006
- Huang, N., & Phillips, C. (2021). When missing nps make double center-embedding sentences acceptable. *Glossa: a journal of general linguistics*, 6(1).
- Huang, Y., & Ferreira, F. (2021). What causes lingering misinterpretations of garden-path sentences: Incorrect syntactic representations or fallible memory processes? *Journal of Memory and Language*, 121, 104288.
- Ivanova-Sullivan, T., Sekerina, I. A., & Lago, S. (2024). Bulgarian clitics are sensitive to number attraction. *Glossa Psycholinguistics*, 3(1).
- Jasinskaja, K., & Šimík, R. (in press). Slavonic free word order. In J. Fellerer & N. Bermel (Eds.), *The Oxford Guide to the Slavonic Languages*. Oxford University Press.
- Jeretič, P. (2023). Exceptionally optional negative concord with turkish neither... nor. *Natural Language & Linguistic Theory*, 41(3), 1147–1200.
- Kamali, B., & Zeijlstra, H. (2024). Negative dependencies in turkish. *Languages*, 9(11), 342.
- Kaup, B., & Dudschig, C. (2020). Understanding negation: Issues in the processing of negation. In *The Oxford Handbook of Negation*. Oxford University Press.
- Kim, K.-k. (2001). *Korean negation and the licensing condition on negative polarity items* (Unpublished doctoral dissertation). University of Hawai'i at Manoa.
- Krifka, M. (1995). The semantics and pragmatics of polarity items. *Linguistic analysis*, 25(3-4), 209–257.
- Kuhn, J. D. (2019). Negative concord as a marker of empty discourse referents. In *Semantics and linguistic theory* (pp. 341–350).

- Kuhn, J. D. (2022). The dynamics of negative concord. *Linguistics and Philosophy*, 45(1), 153–198. doi: 10.1007/s10988-020-09318-3
- Kush, D., Wu, M., & Khurana, A. (2026). Negative polarity illusions are robust with both ‘ever’ and ‘any’ (when linear position is held constant). *Cognition*, 266, 106295.
- Kuznetsova, A., Brockhoff, P. B., Christensen, R. H., & others. (2017). lmerTest package: tests in linear mixed effects models. *Journal of statistical software*, 82(13), 1–26. doi: <https://doi.org/10.18637/jss.v082.i13>
- Kwon, N., & Sturt, P. (2016). Attraction effects in honorific agreement in Korean. *Frontiers in psychology*, 7, 1302.
- Lacina, R. (2025). The negative concord illusion: An acceptability study with Czech neg-words. *Linguistics Vanguard*.
- Lacina, R., & Chromý, J. (2022). No agreement attraction facilitation observed in Czech: Not even syncretism helps. In *Proceedings of the Annual Meeting of the Cognitive Science Society* (Vol. 44, pp. 2423–2430). Retrieved from <https://escholarship.org/uc/item/2vq4g622>
- Lacina, R., & Dotlačil, J. (2024). Grammaticality illusions in Czech: A speeded acceptability study of agreement attraction. In *Proceedings of the annual meeting of the cognitive science society* (Vol. 46).
- Lacina, R., Laurinavichyute, A., & Chromý, J. (2025). Only case-syncretic nouns attract: Czech and Slovak gender agreement. *Journal of Memory and Language*.
- Ladusaw, W. A. (1992). Expressing negation. In *Semantics and linguistic theory* (pp. 237–260).
- Lago, S., Gračanin-Yukse, M., Şafak, D. F., Demir, O., Kırkıcı, B., & Felser, C. (2019). Straight from the horse’s mouth: agreement attraction effects with Turkish possessors. *Linguistic Approaches to Bilingualism*, 9(3), 398–426. doi: 10.1075/lab.17019.lag
- Lago, S., Shalom, D. E., Sigman, M., Lau, E. F., & Phillips, C. (2015). Agreement attraction in Spanish comprehension. *Journal of Memory and Language*, 82, 133–149. doi: 10.1016/j.jml.2015.02.002
- Lewis, R. L., & Vasishth, S. (2005). An activation-based model of sentence processing as skilled memory retrieval. *Cognitive science*, 29(3), 375–419. doi: 10.1207/s15516709cog0000_25
- Liu, H. (2010). Dependency direction as a means of word-order typology: A method based on dependency treebanks. *Lingua*, 120(6), 1567–1578.
- Logačev, P., & Vasishth, S. (2016). A multiple-channel model of task-dependent ambiguity resolution in sentence comprehension. *Cognitive Science*, 40(2), 266–298. doi: 10.1111/cogs.12228

- Mendia, J. A., Poole, E., & Dillon, B. (2018). Spurious npi licensing and exhaustification. In *Semantics and linguistic theory* (pp. 233–250).
- Muller, H. (2022). *What could go wrong? linguistic illusions and incremental interpretation* (Unpublished doctoral dissertation). University of Maryland, College Park.
- Muller, H., & Phillips, C. (2020, 03). Negative Polarity Illusions. In *The Oxford Handbook of Negation*. Oxford University Press. Retrieved from <https://doi.org/10.1093/oxfordhb/9780198830528.013.42> doi: 10.1093/oxfordhb/9780198830528.013.42
- Ng, A., & Husband, M. (2017). Interference effects across the at-issue/not-at-issue divide: Agreement and NPI licensing. In *The 30th Annual CUNY Human Sentence Processing Conference*.
- O'Connor, E. (2015). *Comparative illusions at the syntax-semantics interface* (Unpublished doctoral dissertation). University of Southern California.
- O'Connor, E., Pancheva, R., & Kaiser, E. (2013). Evidence for online repair of escher sentences. In *Proceedings of sinn und bedeutung* (Vol. 17, pp. 363–380).
- Orth, W. J. (2023). *Active assignment of quantifier scope guides language processing* (Unpublished doctoral dissertation). Northwestern University.
- Orth, W. J., Sloggett, S., & Yoshida, M. (2025). Positive polarity items: an illusion of ungrammaticality. *Language, Cognition and Neuroscience*, 0(0), 1–25. Retrieved from <https://doi.org/10.1080/23273798.2025.2453183> doi: 10.1080/23273798.2025.2453183
- Orth, W. J., Yoshida, M., & Sloggett, S. (2021). Negative polarity item (npi) illusion is a quantification phenomenon. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 47(6), 906.
- Ovalle, L. A., & Guerzoni, E. (2004). Double negatives, negative concord and metalinguistic negation. In A. P. Mary Androni Erin Debenport & K. Yoshimura (Eds.), *Proceedings from the main session of the thirty-eighth meeting of the Chicago Linguistic Society: Volume 38-1* (pp. 15–31). Chicago, IL: Chicago Linguistic Society Publications.
- Paape, D. (2024). How do linguistic illusions arise? rational inference and good-enough processing as competing latent processes within individuals. *Language, Cognition and Neuroscience*, 39(10), 1334–1365. Retrieved from <https://doi.org/10.1080/23273798.2024.2387226> doi: 10.1080/23273798.2024.2387226
- Paape, D., Vasishth, S., & Engbert, R. (2021, 07). Does local coherence lead to targeted regressions and illusions of grammaticality? *Open Mind*, 5, 42–58. Retrieved from https://doi.org/10.1162/opmi_a_00041 doi: 10.1162/opmi_a_00041

- Pañeda, C., & Lago, S. (2024). The missing vp illusion in spanish: Assessing the role of language statistics and working memory. *Open Mind*, 8, 42–66.
- Parker, D. (2025a). Comprehending negation. In *Perspectives on Negation: Views from Across the Language Sciences*. Walter de Gruyter.
- Parker, D. (2025b). *Linguistic Illusions: A Case Study on Agreement Attraction*. Cambridge University Press.
- Parker, D., & An, A. (2018). Not all phrases are equally attractive: Experimental evidence for selective agreement attraction effects. *Frontiers in psychology*, 9, 1566. doi: 10.3389/fpsyg.2018.01566
- Parker, D., & Phillips, C. (2016). Negative polarity illusions and the format of hierarchical encodings in memory. *Cognition*, 157, 321–339.
- Paspali, A., & Marinis, T. (2020). Gender agreement attraction in Greek comprehension. *Frontiers in psychology*, 11, 717. doi: 10.3389/fpsyg.2020.00717
- Pasquereau, J., Dillon, B., & Frazier, L. (2024). Quantification at a distance and grammatical illusions in french. *Syntax*, 27(1), 1–47.
- Patson, N. D., & Husband, E. M. (2016). Misinterpretations in agreement and agreement attraction. *Quarterly Journal of Experimental Psychology*, 69(5), 950–971. Retrieved from <https://doi.org/10.1080/17470218.2014.992445> (PMID: 25624171) doi: 10.1080/17470218.2014.992445
- Penka, D. (2011). *Negative indefinites*. Oxford: Oxford University Press.
- Phillips, C., Wagers, M., & Lau, E. (2011). Grammatical illusions and selective fallibility in real-time language comprehension. In *Experiments at the Interfaces* (pp. 147–180). Brill.
- Potter, M. C. (2018). Rapid serial visual presentation (rsvp): A method for studying language processing. In *New methods in reading comprehension research* (pp. 91–118). Routledge.
- R Core Team. (2023). *R: A language and environment for statistical computing* (manual). Vienna, Austria. Retrieved from <https://www.R-project.org/>
- Royer, A. J. (2021). *The role of prosodic phrasing on the acceptability of agreement attraction in english*. University of California, Los Angeles.
- Runner, J., Sussman, R., & Tanenhaus, M. (2006). Processing reflexives and pronouns in picture noun phrase. *Cognitive Science*, 30(2), 193–241.
- Schwab, J. (2023). Lexical variation in np_i illusions—a case study of german jemals ‘ever’ and so recht ‘really’. *Glossa Psycholinguistics*, 2(1).

- Schwab, J., & Liu, M. (2022). Attenuating npis in indicative and counterfactual conditionals. In *Proceedings of Sinn und Bedeutung* (Vol. 26, pp. 772–789).
- Shimoyama, J. (2011). Japanese indeterminate negative polarity items and their scope. *Journal of Semantics*, 28(4), 413–450.
- Siewierska, A., & Uhlířová, L. (1998). An overview of word order in Slavic languages. In A. Siewierska (Ed.), *Constituent order in the languages of Europe* (pp. 105–149). Berlin/New York: De Gruyter.
- Slattery, T. J., Sturt, P., Christianson, K., Yoshida, M., & Ferreira, F. (2013). Lingerin misinterpretations of garden path sentences arise from competing syntactic representations. *Journal of Memory and Language*, 69(2), 104–120.
- Slioussar, N. (2018). Forms and features: The role of syncretism in number agreement attraction. *Journal of Memory and Language*, 101, 51–63. doi: 10.1016/j.jml.2018.03.006
- Slioussar, N., & Malko, A. (2016). Gender agreement attraction in Russian: production and comprehension evidence. *Frontiers in psychology*, 7, 1651. doi: 10.3389/fpsyg.2016.01651
- Sturt, P., & Kwon, N. (2024). Agreement attraction in comprehension: Do active dependencies and distractor position play a role? *Language, Cognition and Neuroscience*, 39(3), 279–301. Retrieved from <https://doi.org/10.1080/23273798.2023.2269282> doi: 10.1080/23273798.2023.2269282
- Su, N., & Aparicio, H. (2022). Npi licensing and intrusion effects in japanese. In *Proceedings of the Chicago Linguistic Society* (Vol. 58, pp. 443–454).
- Szabolcsi, A. (2004). Positive polarity–negative polarity. *Natural language & linguistic theory*, 22(2), 409–452.
- Thorsen, W. M. (2025). Negativ polaritet og illusorisk acceptabilitet i dansk og engelsk. *Moderne om Udforskningen af Dansk Sprog (MUDS)*(20), 389–411.
- Todorović, D. (2020). What are visual illusions? *Perception*, 49(11), 1128–1199.
- Tucker, M. A., Idrissi, A., & Almeida, D. (2015). Representing number in the real-time processing of agreement: Self-paced reading evidence from Arabic. *Frontiers in psychology*, 6, 347. doi: 10.3389/fpsyg.2015.00347
- Tucker, M. A., Idrissi, A., & Almeida, D. (2021). Attraction effects for verbal gender and number are similar but not identical: Self-paced reading evidence from Modern Standard Arabic. *Frontiers in psychology*, 3774. doi: 10.3389/fpsyg.2020.586464

- Türk, U., Atmaca, F., Özateş, Ş. B., Berk, G., Bedir, S. T., Köksal, A., ... Özgür, A. (2022). Resources for turkish dependency parsing: Introducing the boun treebank and the boat annotation tool. *Language Resources and Evaluation*, 56(1), 259–307.
- Turk, U., & Logačev, P. (2021). Agreement Attraction in Turkish: The Case of Genitive Attractors. *PsyArXiv*. doi: 10.31234/osf.io/5rmvu
- Vasishth, S., Brüssow, S., Lewis, R. L., & Drenhaus, H. (2008). Processing polarity: How the ungrammatical intrudes on the grammatical. *Cognitive Science*, 32(4), 685–712. Retrieved from <https://onlinelibrary.wiley.com/doi/abs/10.1080/03640210802066865> doi: <https://doi.org/10.1080/03640210802066865>
- von Fintel, K. (1999). NPI licensing, Strawson entailment, and context dependency. *Journal of Semantics*, 16(2), 97–148. doi: 10.1093/jos/16.2.97
- Wagers, M., Lau, E., & Phillips, C. (2009). Agreement attraction in comprehension: Representations and processes. *Journal of Memory and Language*, 61(2), 206–237. doi: 10.1016/j.jml.2009.04.002
- Wellwood, A., Pancheva, R., Hacquard, V., & Phillips, C. (2018). The anatomy of a comparative illusion. *Journal of Semantics*, 35(3), 543–583.
- Xiang, M., Dillon, B., & Phillips, C. (2009). Illusory licensing effects across dependency types: Erp evidence. *Brain and Language*, 108(1), 40–55.
- Xiang, M., Grove, J., & Giannakidou, A. (2013). Dependency-dependent interference: Npi interference, agreement attraction, and global pragmatic inferences. *Frontiers in psychology*, 4, 708.
- Yanilmaz, A., & Drury, J. E. (2018). Prospective npi licensing and intrusion in turkish. *Language, Cognition and Neuroscience*, 33(1), 111–138.
- Yun, J., Lee, S.-Y., & Drury, J. E. (2018). Negative polarity illusion in korean. *Proceedings of WAFL*, 13.
- Zehr, J., & Schwarz, F. (2018). PennController for internet based experiments (IBEX). doi: 10.17605/OSF.IO/MD832
- Zeijlstra, H. (2004). *Sentential negation and negative concord* (Unpublished doctoral dissertation). Utrecht University.
- Zeijlstra, H. (2008). Negative concord is syntactic agreement. *Ms., University of Amsterdam*, 5, 113.
- Zeijlstra, H. (2022). *Negation and negative dependencies* (Vol. 80). Oxford University Press.

Zhang, Y. (2024). *The rational processing of language illusions* (Unpublished doctoral dissertation).
Harvard University.